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Foreword

This Technical Report has been produced by the 3rd Generation Partnership Project (3GPP).

The present document is part of a multi-part TR as described below:

- Part 1: LI for IMS based services. See TR 33.929-1 [6].
- Part 2: LI for IMS based STIR/SHAKEN. See TR 33.929-2 [7].
- Part 3: LI for messaging services. See TR 33.929-3 [8].
- Part 4: LI for data in 5G core. See TR 33.929-4 [9].
- Part 5: LI for ID association caching (present document).
- Part 6: LI for IMS based RCS. See TR 33.929-6 [10].
- Part 7: LI location acquisition capabilities. See TR 33.929-7 [11].
- Part 8: LI for MMS. See TR 33.929-8 [12].
- Part 9: LI for PTC. See TR 33.929-9 [13].

In Part 1, the illustrations of LI for IMS-based services focus on the LI aspects of IMS sessions which include the architecture topologies and the call flows covering basic sessions, redirected sessions, target non-local ID, conferencing, roaming (local break-out and home-routed roaming).

In Part 2, the illustrations of LI for IMS based STIR/SHAKEN focus on various STIR/SHAKEN related LI reporting scenarios.

In Part 3, the illustrations of LI for messaging services focus on SMS over NAS and SMS over IP.

In Part 4, the illustrations of LI of data in 5G core focus on the LI aspects of data interception focussing on the PDU session related events including the handover scenarios.

In Part 5 (present document), the illustrations of LI for ID association caching focus on the capabilities used to provide the temporary identity to permanent identity (and vice-versa) associations known to the CSP network. The illustrations include the architectural concepts and few examples of use-cases encountering various timing scenarios.

In Part 6, the illustrations of LI for IMS based RCS focus on the architecture topologies and the call flows when the RCS service is offered by the CSP or by a Third Party Provider.

In Part 7, the illustrations of LI location acquisition capabilities focus on the conceptual overview and the flow diagrams for location reporting, LALS and location acquisition.

In Part 8, the illustrations of LI for MMS focus on the conceptual overview and the flow diagrams for MMS.

In Part 9, the illustrations of LI for PTC focus on the conceptual overview and the flow diagrams for MCPTT and PoC, the two commonly referred to as Push to Talk over Cellular (PTC).

Introduction

Unlike the previous generation of LI technical specifications, the latest LI technical specifications (TS 33.126 [2], TS 33.127 [3], TS 33.128 [4]) contain only the normative part of the requirements. However, the implementers may need additional information such as architectural diagrams, conceptual scenarios, flow-diagrams and additional background information to better understand the requirements in those specifications. The present document collects the relevant informative annexes from the previous generation of LI technical specifications and then expands them to include the similar level of details as for latest technologies such as 5G.

1 Scope

The present document provides architectural diagrams, conceptual scenarios, flow-diagrams, examples, and other background information which can be useful to implement the LI functions defined in TS 33.126 [2], TS 33.127 [3], TS 33.128 [4] and TR 33.928 [5].

The present document that covers illustrations of ID Association Caching focus on the capabilities used to provide the temporary identity to permanent identity (and vice-versa) associations known to the CSP network. The illustrations include the architectural concepts and few examples of use-cases encountering various timing scenarios.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 33.126: "Lawful Interception requirements".

[3] 3GPP TS 33.127: "Lawful Interception (LI) architecture and functions".

[4] 3GPP TS 33.128: "Protocol and procedures for Lawful Interception (LI)".

[5] 3GPP TR 33.928: "ADMF logic for provisioning Lawful Interception (LI)".

[6] 3GPP TR 33.929-1: "Lawful Interception (LI) implementation guidance; LI for IMS-based services".

[7] 3GPP TR 33.929-2: "Lawful Interception (LI) implementation guidance; LI for IMS based STIR/SHAKEN".

[8] 3GPP TR 33.929-3: "Lawful Interception (LI) implementation guidance; LI for messaging services".

[9] 3GPP TR 33.929-4: "Lawful Interception (LI) implementation guidance; LI for data in 5G core".

[10] 3GPP TR 33.929-6: "Lawful Interception (LI) implementation guidance; LI for IMS based RCS".

[11] 3GPP TR 33.929-7: "Lawful Interception (LI) implementation guidance; LI location acquisition capabilities".

[12] 3GPP TR 33.929-8: "Lawful Interception (LI) implementation guidance; LI for MMS".

[13] 3GPP TR 33.929-9: "Lawful Interception (LI) implementation guidance; LI for PTC".

3 Definitions of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in 3GPP TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [1].

3.2 Symbols

For the purposes of the present document, the following symbols apply:

None.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [1].

5G-GUTI	5G Globally Unique Identifier
5G-S-TMSI	5G-Serving-Temporary Mobile Subscription Identifier
CSP	Communication Service Provider
AMF	Access and Mobility Management
ICF	Identity Caching Function
ID	Identity
IEF	Identity Event Function
IQF	Identity Query Function
LALS	Lawful Access Location Services
LEA	Law Enforcement Agency
LI	Lawful Interception
LI_HIQR	Lawful Interception Handover Interface Query Response
LI_XER	Lawful Interception Internal Interface Event Record
LI_XQR	Lawful Interception Internal Interface Query Response
MCC	Mobile Country Code
MCPTT	Mission Critical Push To Talk
MNC	Mobile Network Code
NCGI	NR Cell Global Identity
PEI	Permanent Equipment Identifier
PoC	Push to talk over Cellular
PTC	Push to Talk over Cellular
SUCI	Subscriber Concealed Identifier
SUPI	Subscriber Permanent Identifier
TA	Tracking Area
TAC	Tracking Area Code
TA-ID	Tracking Area Identifier
TAI	Tracking Area Identity
TMSI	Temporary Mobile Subscription Identifier
TR	Technical Report
TS	Technical Specification
UE	User Equipment

4 ID Association Caching

4.1 Introduction

With the advent of 5G, the permanent IDs are not freely circulated within the network or on the radio. LEAs have identified requirements for the CSP to be able to determine the permanent IDs from the non-permanent IDs or vice-versa. The secured and controlled real-time process implemented within the CSP network domain to support the above capabilities as specified in TS 33.127 [3] and TS 33.128 [4] is referred to as ID Association Caching.

An overview of the architecture as defined in TS 33.127 [3] for the ID Association Caching function is depicted below in figure 4.1-1.

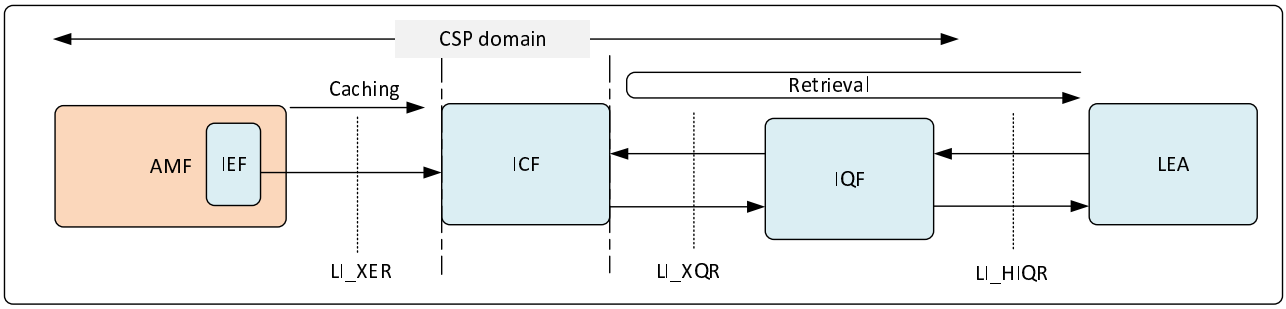


Figure 4.1-1: ID Association Caching - architecture

As shown in figure 4.1-1, the concept of ID Association Caching includes two functions:

- Caching.
- Retrieval.

The Caching function that applies to all registered UEs is handled within the CSP domain. The Retrieval function that applies to a specific UE based on a warrant is handled with a Request/Response step involving the LEAs and the CSP.

NOTE: In this clause, an association between a permanent ID and the non-permanent ID is referred to as ID association.

4.2 IDs

ID Association Caching references the following IDs:

Non-permanent IDs:

- SUCI.
- 5G-GUTI.
- 5G-S-TMSI.

Permanent IDs:

- SUPI.
- PEI.
- GPSI.

5G-S-TMSI is always part of the 5G-GUTI as shown in figure 4.2-1 below.

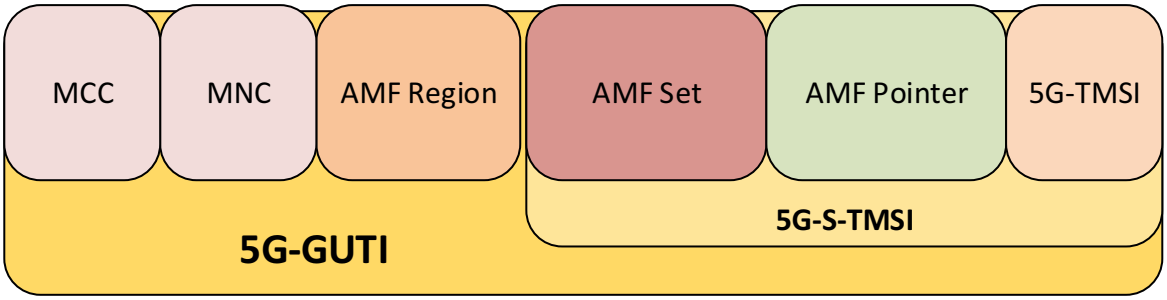


Figure 4.2-1: Structure of 5G-GUTI and 5G-S-TMSI

The LEAs may include the SUCI or the 5G-S-TMSI or the 5G-GUTI in the Request to retrieve the associated SUPI and PEI. As a different scenario, the LEAs may instead include the SUPI in the Request to retrieve the associated 5G-GUTI.

4.3 Caching Principles

4.3.1 Background

When a UE registers to the network or requests for a service, the AMF present in the 5GC assigns a new 5G-GUTI to that UE, or more precisely, to the SUPI associated with that UE.

Whenever a new 5G-GUTI is successfully assigned to the SUPI, the previous 5G-GUTI is presumed to be disassociated from that SUPI.

4.3.2 Caching

Whenever the AMF assigns a new 5G-GUTI to a SUPI, the IEF present within the AMF caches 5G-GUTI to SUPI association (referred to as ID association) at the ICF.

The SUCI (when known), GPSI (when known) and the PEI (when known) may also be tagged along with the 5G-GUTI to SUPI association. The Cell-Id and TA-ID are also cached. The Association Start Time (see clause 4.3.4.9) is also cached.

For caching the ID association, the IEF present in the AMF sends the IEF Association Record message to the ICF over the LI_XER interface as defined in TS 33.128 [4].

The ICF, upon receiving the IEF Association Record message, caches the new ID association with the all the information that it receives within the message. If the previously cached ID association for that SUPI contains the PEI, then the ICF transfers that PEI to the new ID association unless the new ID association includes the SUCI. This is because a new ID association for the same SUPI not including a SUCI is presumed to be for the same PEI. However, when a new ID association for the same SUPI that includes the SUCI is received, it is possible that the PEI may have changed and was not reported.

If the previously cached ID association for that SUPI contains the GPSI, then the ICF transfers that GPSI to the new ID association unless the new ID association includes the GPSI.

IEF present in the AMF also sends an IEF Association Record message when it detects that the GPSI is changed. The ICF, upon receiving the IEF Association Record message, caches the new ID association with the all the information it receives within the message.

4.3.3 De-caching

The term "de-caching" is used to refer to the process of deleting the ID association from the cache after a short time-period, which in turn, is referred to as de-caching period.

The de-caching process begins with the ICF marking the cached ID association for deletion, starting the de-caching period timer, and then deleting the ID association from the cache when the de-caching period timer expires. The Association End Time (see clause 4.3.4.9) is recorded within the cached ID association during the de-caching period.

There are two ways an ID association within the ICF can be de-cached:

- When a new IEF Association Record message is received, the previous ID association is de-cached.
- When the ICF receives an IEF Disassociation Record message from the IEF.

In the first case, the ICF upon receiving an IEF Association Record message to cache a new ID association, automatically begins the de-caching of the previously cached ID association for that SUPI.

In the second case, the IEF upon detecting that the AMF has disassociated a 5G-GUTI from a SUPI without assigning a new 5G-GUTI, sends an IEF Disassociation Record message to the ICF over the LI_XER interface (as defined in TS 33.128 [4]). The ICF, upon receiving IEF Disassociation Record message, begins the de-caching of ID association for that SUPI.

4.3.4 Summary of the terms used within the caching principles

4.3.4.1 Illustration

Figure 4.3.4.1-1 illustrates the concepts behind some of the terms used in this clause.

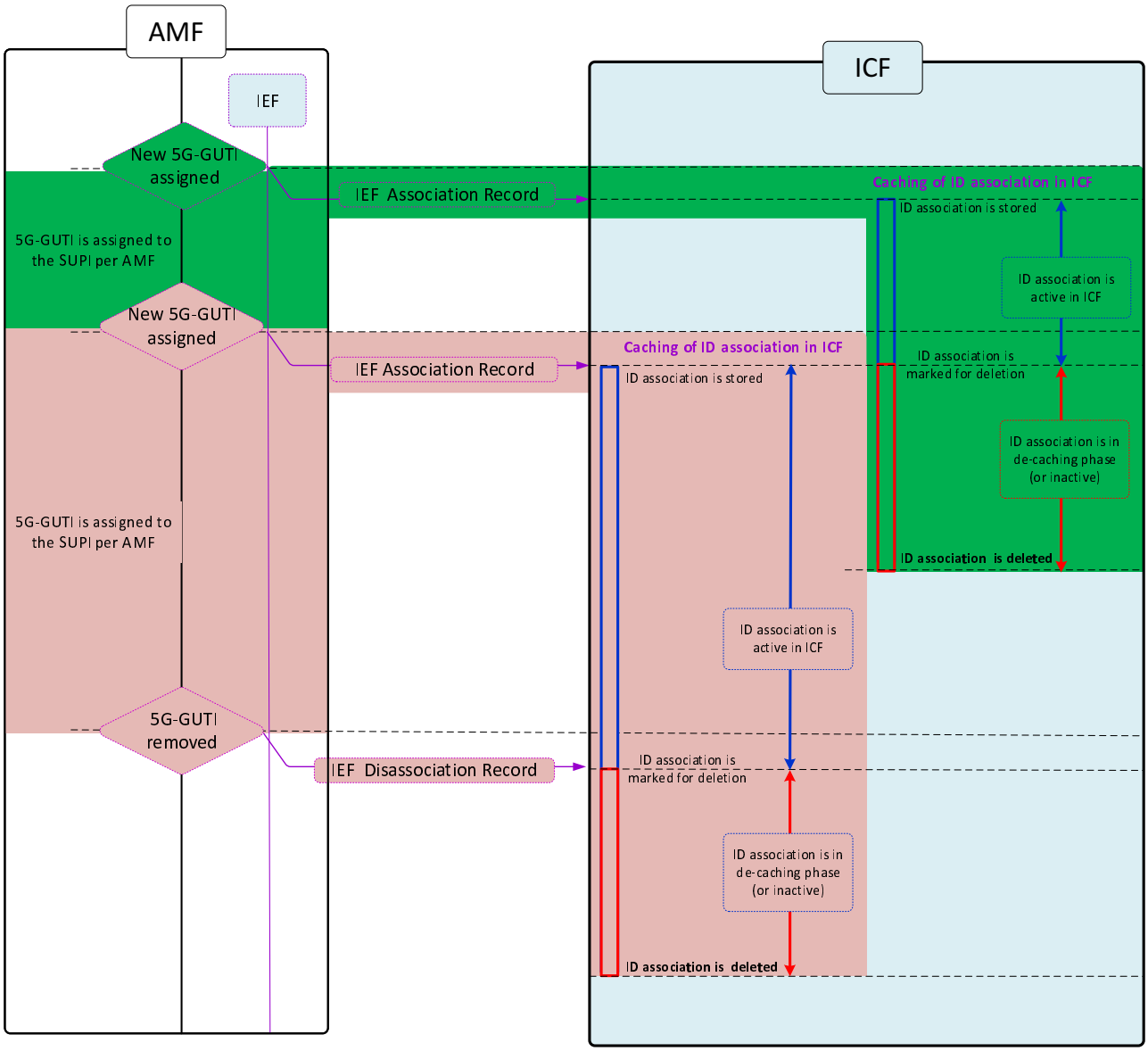


Figure 4.3.4.1-1: Terms used in within caching principles

The following clauses provide a summary of the terms shown in figure 4.3.4.1-1.

4.3.4.2 ID association

An association between the permanent ID and non-permanent ID is cached in the ICF. The ICF receives the ID association within the IEF Identity Association Record message from the IEF.

4.3.4.3 Caching

The process of the IEF sending the IEF Identity Association Record to the ICF and the ICF caching the information received in the IEF Identity Association Record message in its memory.

4.3.4.4 De-caching

The process of marking an ID association for deletion, starting the de-caching period timer, and then deleting the cached ID association when the de-caching period timer expires.

4.3.4.5 De-caching period

The period from the time an ID association is marked for deletion to the time the ID association is removed. This can also be said that it is the time-period of de-caching process.

4.3.4.6 Active ID association

The cached ID association before its de-caching begins (i.e. marked for deletion) is referred to as active ID association in the ICF.

4.3.4.7 ID association in a de-caching phase

The ID association, still in the ICF memory, but is part of a de-caching process is referred to as ID association in a de-caching phase.

4.3.4.8 Inactive ID association

The ID association, still in the ICF memory, but in a de-caching phase can also be referred to as inactive ID association in the ICF.

4.3.4.9 Association Start Time and Association End Time

The time at which the 5G-GUTI is assigned to the SUPI (in the AMF) is referred to as Association Start Time. The time at which the 5G-GUTI is presumed to be disassociated from a SUPI is referred to as Association End Time.

When an ID association is marked for deletion in the ICF due to the reception of a new ID association, the Association Start Time of a new cached ID association is same as the Association End Time of a previously cached ID association. The Association Start Time is included within the IEF Association Record message that the IEF sends to the ICF.

As described in 4.3.3, the 5G-GUTI disassociation from a SUPI in the AMF may also happen without a new 5G-GUTI association (e.g. UE de-registering from the network). In other words, the Association End Time of a cached ID association does not always be the Association Start Time of a new cached ID association. The Association End Time is included within IEF Disassociation Record message that IEF sends to the ICF.

An illustration of timing concepts is shown in figure 4.3.4.9-1 below.

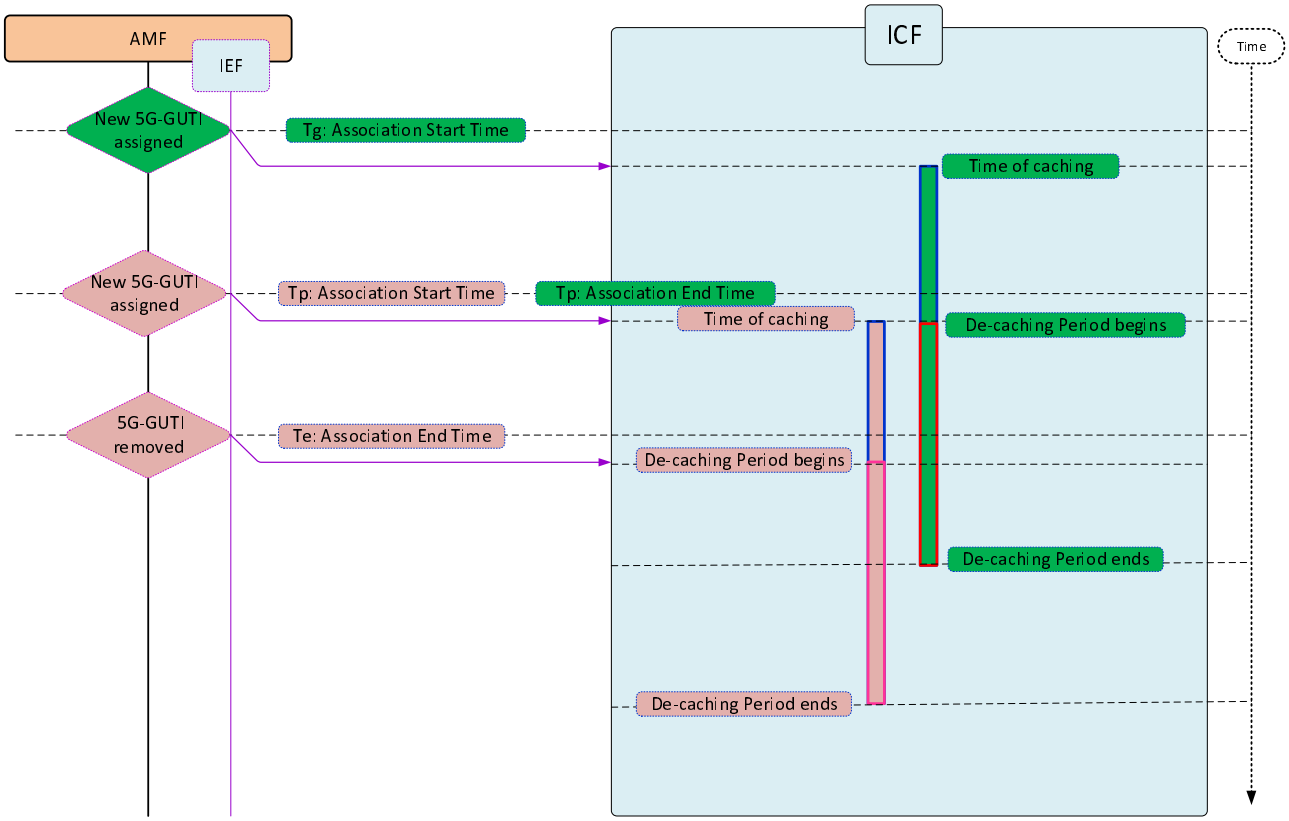


Figure 4.3.4.9-1: Timing concepts associated caching principles

As shown in figure 4.3.4.9-1, Tg and Tp are the Association Start Time for the two shown ID associations cached in the ICF. The Tp is the Association End Time for the first ID association. Te is the Association End Time for the second ID association.

NOTE: The Association Start Time is not the time when a new ID association is cached in the ICF. The Association End Time is neither the time when the cached ID is marked for removal (i.e. de-caching period begins) in the ICF nor the time when the cached ID is removed from the ICF.

4.3.5 Illustrations of caching and de-caching principles

4.3.5.1 Multiple ID associations for a SUPI

When a new IEF Association Record message is received, the ICF de-caches (see clause 4.3.3) the previous ID association. During the de-caching phase of the previous ID association (s), it is possible that ICF may have new ID associations for that SUPI. In other words, at any given time, multiple ID associations for a SUPI may be present within the ICF. But only one will be the active ID association at any given time.

This is illustrated in figure 4.3.5.1-1 below.

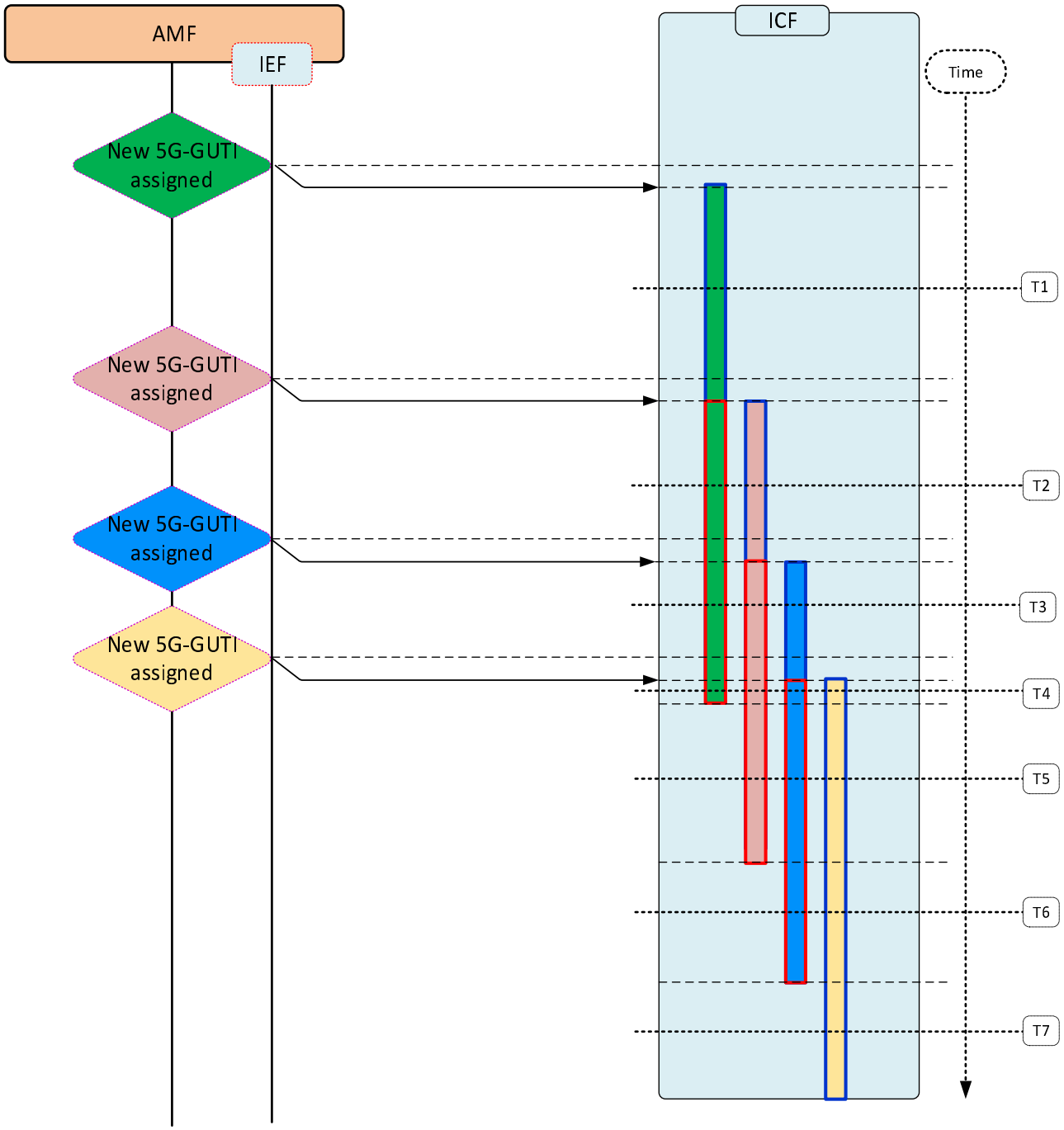


Figure 4.3.5.1-1: Multiple ID associations in the ICF for the same SUPI

NOTE: The focus of the above figure is on the ID association caching and de-caching. That is why the arrows between the IEF and ICF do not show the message names.

As shown in figure 4.3.5.1-1, at time T2, T3, T4, T5 and T6 more than one ID associations are in the ICF for the same SUPI. But only one ID association is active at all those times.

4.3.5.2 5G-GUTI in multiple ID associations

A 5G-GUTI that is no longer in use may be re-assigned to a new UE (i.e. to a new SUPI) over time by the AMF.

When a new IEF Association Record message is received, the ICF de-caches (see clause 4.3.3) the previous ID association. ,. However, since the AMF that re-assigns the 5G-GUTI to a new SUPI is unaware of the ICF existence, or its handling of the ID association de-caching, it is possible that the same 5G-GUTI is present in multiple ID

associations, at a given time within the ICF. This can be the case when the de-caching period is longer than the time an AMF takes to re-assign a freed 5G-GUTI to a new SUPI.

NOTE 1: If the time an AMF takes to assign a freed 5G-GUTI is always greater than de-caching period (which in most implementations can be the case), then the scenario depicted here can only be a theoretical possibility. Even then, the ICF implementations should not dismiss such a possibility.

This is illustrated in figure 4.3.5.2-1 below.

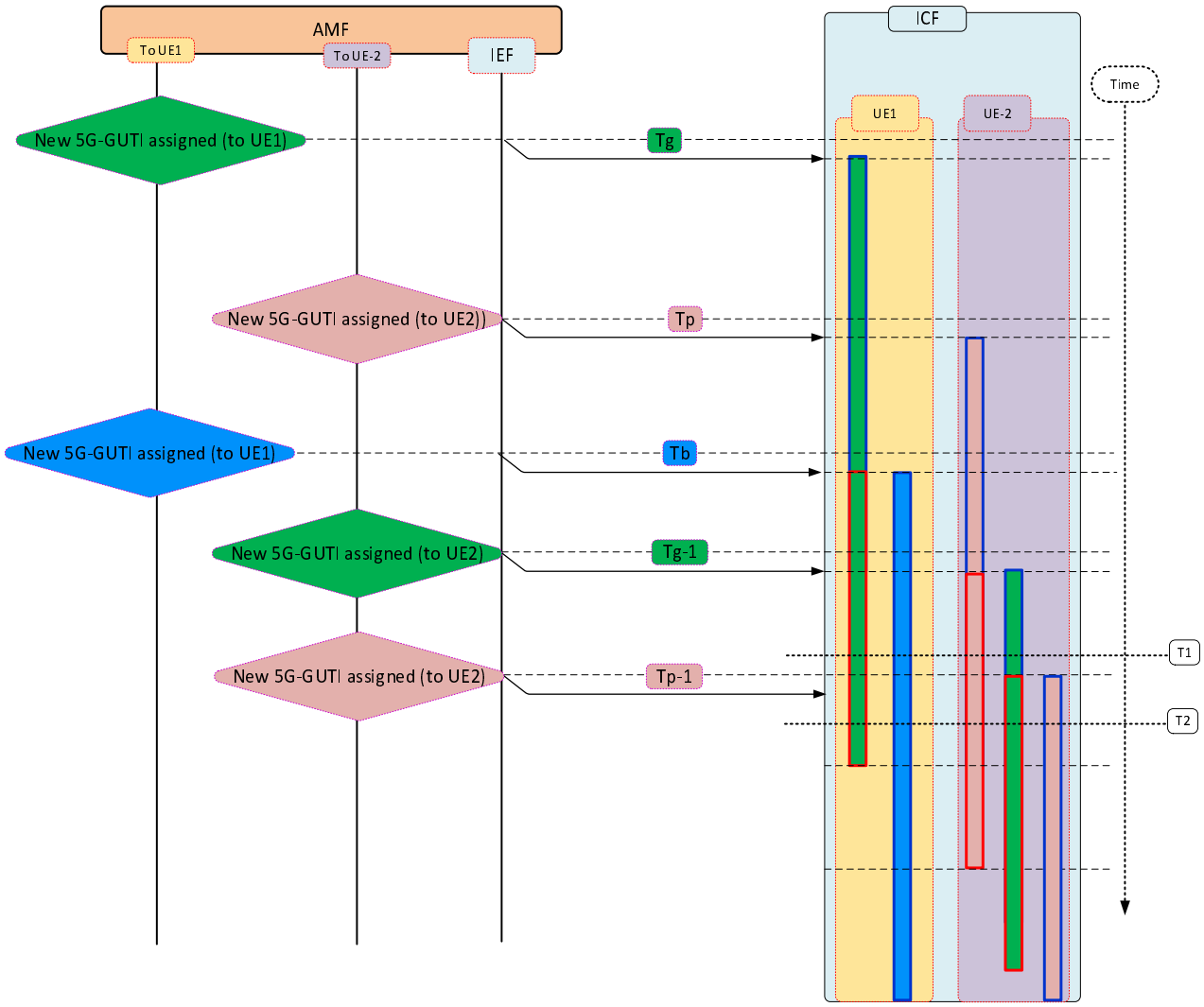


Figure 4.3.5.2-1: 5G-GUTI in multiple cached ID associations

NOTE 2: The focus of the above figure is on the ID association caching and de-caching. That is why the arrows between the IEF and ICF do not show the message names.

Figure 4.3.5.2-1 shows the 5G-GUTI assignments for two UEs. As shown, at time T1, the same 5G-GUTI is in ID associations within the ICF (i.e. two different SUPIs), one is in the de-caching phase and the other one is active.

NOTE 3: Even though the illustration shows two different UEs, in theory, a freed 5G-GUTI can be re-assigned by the AMF to the same SUPI later, while a previous ID association with that 5G-GUTI for the SUPI is still in the ICF (in a de-caching phase). This again is a theoretical possibility, however, the ICF implementations should not dismiss the possibility.

4.3.5.3 ID association caching aspects

4.3.5.3.1 Overview

The following information are mandatory for an IEF Association Record message.

5G-GUTI	SUPI	Time-stamp	TAI	N-CGI	N-CGI Time

The Time-stamp present in the IEF Association Record is the Association Start Time for this 5G-GUTI and the Association End Time for the 5G-GUTI in the previous ID association cached in the ICF for the same SUPI.

The following information, if available, should be included in an IEF Association Record message:

SUCI	PEI	GPSI	5G TAI List

The following information are mandatory for an IEF Disassociation Record message:

5G-GUTI	SUPI	Time-stamp	N-CGI	N-CGI Time

The Time-stamp present in the IEF Dissociation Record is the Association End Time for this ID association cached in the ICF for this 5G-GUTI.

4.3.5.3.2 Illustration 1

Figure 4.3.5.3.2-1 illustrates one example of ID association caching aspects within the ICF.

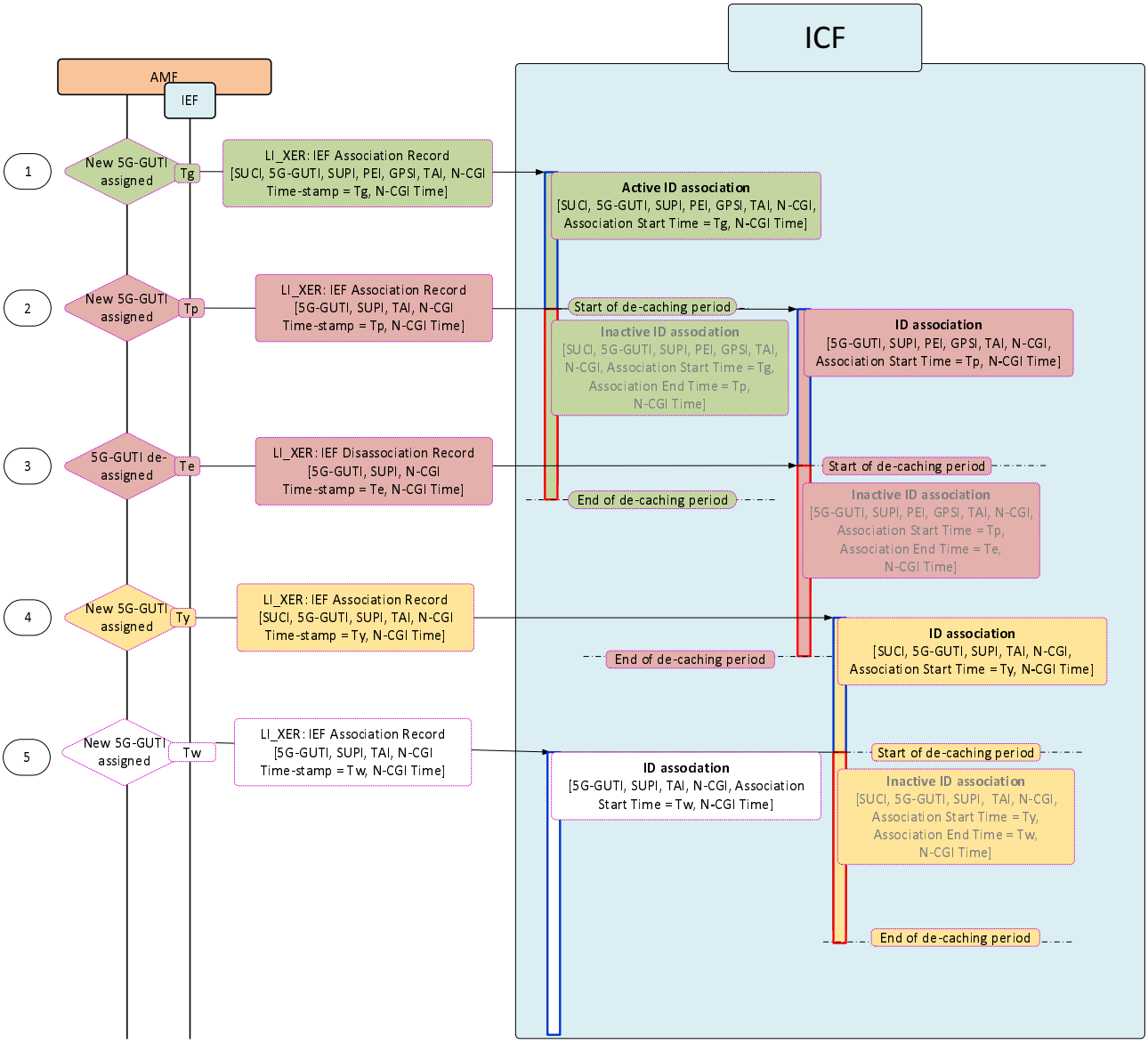


Figure 4.3.5.3.2-1: Caching and de-caching principles

Figure 4.3.5.3.2-1 shows 5 scenarios of the ICF handling of the cached ID associations. Note that in scenario 4, the IEF Association Record has the SUCI but not the PEI. As a result, the subsequent ID associations (scenario 5) will not have the PEI.

1. A 5G-GUTI is assigned to the UE. The IEF sends an IEF Association Record message that contains the SUCI, 5G-GUTI, SUPI, PEI, GPSI, TAI, N-CGI, Time-stamp of Tg and N-CGI-Time. The ICF caches the information as the ID association with Tg as the Association Start Time.

2. A new 5G-GUTI is assigned to the UE. The IEF sends an IEF Association Record message that contains the 5G-GUTI, SUPI, TAI, N-CGI, Time-stamp of Tp and N-CGI-Time. The ICF caches the information as the ID association with Tp as the Association Start Time. The ICF stores the Tp as the Association End Time in the previous ID association and begins the de-caching process for that ID association. The ICF transfers the PEI and GPSI from the previous ID association to the new ID association.

3. The 5G-GUTI is de-assigned for the UE. The IEF sends an IEF Disassociation Record message that contains the 5G-GUTI, SUPI, TAI, N-CGI, Time-stamp of Te and N-CGI-Time. The ICF begins the de-caching process for that ID association with the storing of Te as the Association End Time within the ID association record.

4. A 5G-GUTI is assigned to the UE. The IEF sends an IEF Association Record message that contains the SUCI, 5G-GUTI, SUPI, TAI, N-CGI, Time-stamp of Ty and N-CGI-Time. The ICF caches the information as the ID association with Ty as the Association Start Time. In this example, neither PEI nor GPSI is part of the ID association.

5. A new 5G-GUTI is assigned to the UE. The IEF sends an IEF Association Record message that contains the 5G-GUTI, SUPI, TAI, N-CGI, Time-stamp of Tw and N-CGI-Time. The ICF caches the information as the ID association with Tw as the Association Start Time. The ICF stores the Tw as the Association End Time in the previous ID association and begins the de-caching process for that ID association. In this example, since the previous ID association had neither PEI nor GPSI, the new ID association will also have neither PEI nor GPSI.

4.3.5.3.3 Illustration 2

Figure 4.3.5.3.3-1 illustrates another example of the ID association caching aspects within the ICF.

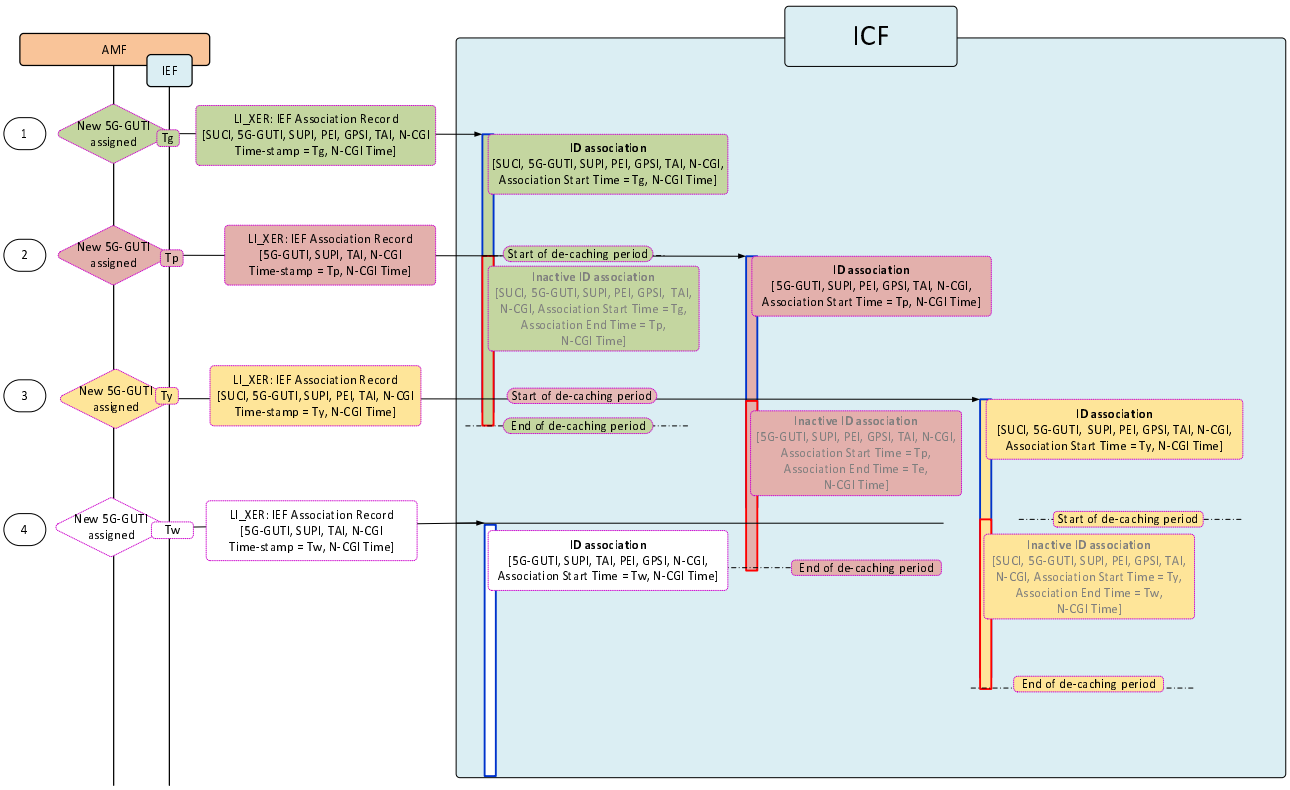


Figure 4.3.5.3.3-1: Caching and de-caching principles

Figure 4.3.5.3.3-1 shows 4 scenarios of the ICF handling of the cached ID associations. Note that in scenario 3, the IEF Association Record has the SUCI and the PEI. Therefore, the subsequent ID associations (scenario 4) will have the PEI.

1. A 5G-GUTI is assigned to the UE. The IEF sends an IEF Association Record message that contains the SUCI, 5G-GUTI, SUPI, PEI, GPSI, TAI, N-CGI, Time-stamp of Tg and N-CGI-Time. The ICF caches the information as the ID association with Tg as the Association Start Time.

2. A new 5G-GUTI is assigned to the UE. The IEF sends an IEF Association Record message that contains the 5G-GUTI, SUPI, TAI, N-CGI, Time-stamp of Tp and N-CGI-Time. The ICF caches the information as the ID association with Tp as the Association Start Time. The ICF stores the Tp as the Association End Time in the previous ID association and begins the de-caching process for that ID association. The ICF transfers the PEI and GPSI from the previous ID association to the new ID association.

3. A new 5G-GUTI is assigned to the UE. The IEF sends an IEF Association Record message that contains the SUCI, 5G-GUTI, SUPI, PEI, TAI, N-CGI, Time-stamp of Ty and N-CGI-Time. The ICF caches the information as the ID association with Ty as the Association Start Time. Since the new ID association has a SUCI, the PEI is not transferred from the previous ID association. However, the PEI (which may be same as the one the previous ID associations, or can be different) received in the IEF Association Record message is cached in the ICF. The ICF transfers the GPSI from the previous ID association to the new ID association.

4. A new 5G-GUTI is assigned to the UE. The IEF sends an IEF Association Record message that contains the 5G-GUTI, SUPI, TAI, N-CGI, Time-stamp of Tw and N-CGI-Time. The ICF caches the information as the ID association with Tw as the Association Start Time. The ICF stores the Tw as the Association End Time in the

previous ID association and begins the de-caching process for that ID association. The ICF transfers the PEI and GPSI from the previous ID association (step 3) to the new ID association.

4.3.6 ID association caching and de-caching scenarios

4.3.6.1 Caching scenarios

A new 5G-GUTI may be assigned to a SUPI when the UE registers with the 5GC or sends a new service request. The multiple possible scenarios are depicted in figure 4.3.6.1-1 below.

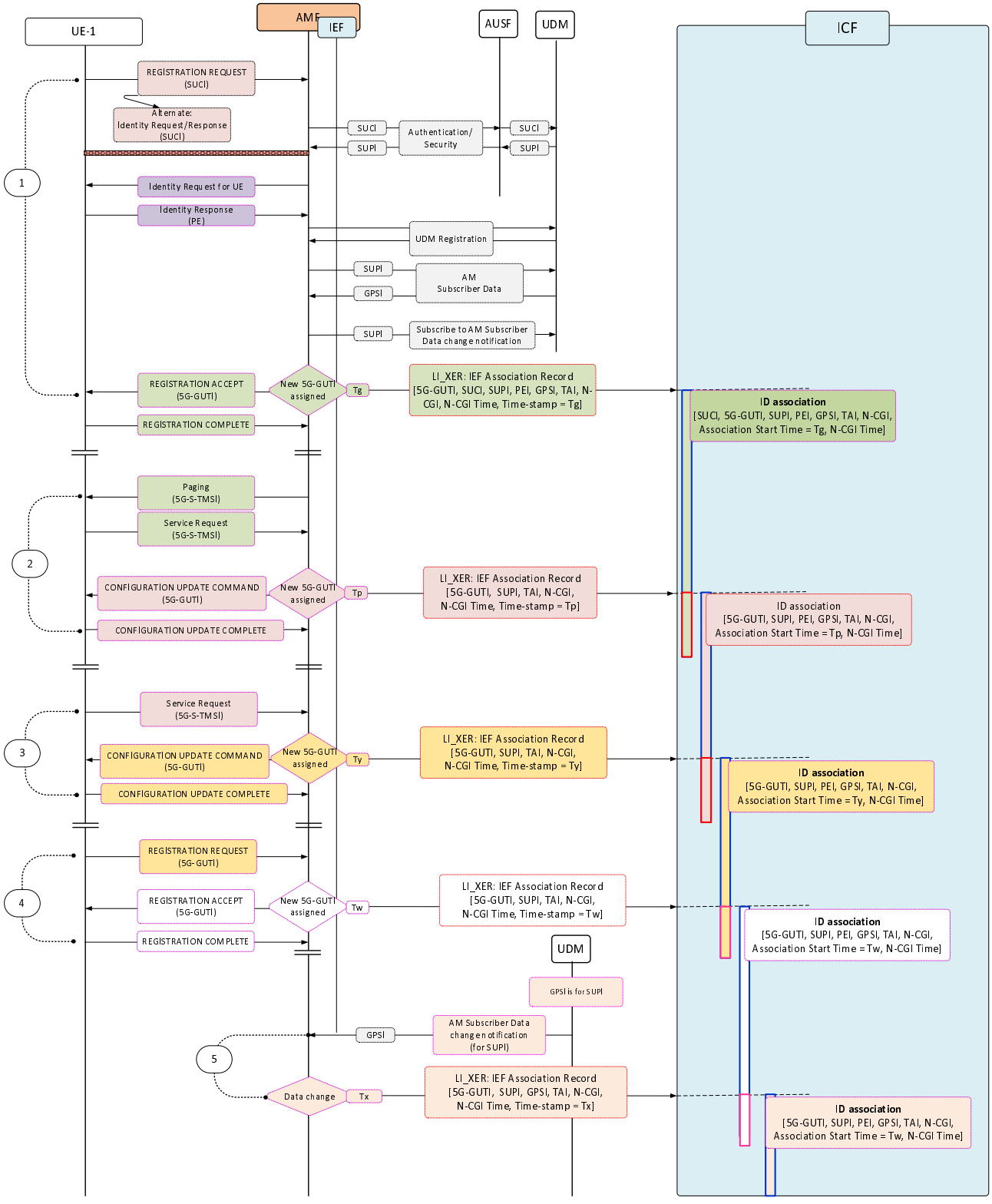


Figure 4.3.6.1-1: Caching scenarios

Figure 4.3.6.1-1 shows 4 scenarios of ID association caching:

Scenario 1: 5G-GUTI in the REGISTRATION ACCEPT to the UE (e.g. power-on scenario):

a) When the UE initially powers on and sends a REGISTRATION REQUEST message to the AMF, the REGISTRATION REQUEST may contain the SUCI or the AMF may request the UE for the same using Identity Request/Response procedures.

b) The AMF then performs the authentication/security checks for the SUCI with the AUSF/UDM and gets the SUPI associated with the SUCI from the AUSF/UDM.

c) The AMF gets the PEI from the UE using Identity Request/Response procedures.

d) After registering with the UDM, the AMF assigns a 5G-GUTI to the UE and sends the same to the UE in the REGISTRATION ACCEPT message. Before that, the AMF may get the Access Mobility (AM) subscriber Data from the UDM which would include the GPSI and then may subscribe to AM Subscriber Data change notification to the UDM. The illustration shows these two steps.

e) The IEF present in the AMF detects the assignment of a new 5G-GUTI and sends the IEF Association Record message to the ICF that includes the 5G-GUTI, SUCI, SUPI, PEI, GPSI, Time-stamp = Tg, N-CGI, TAI, N-CGI Time (see clause 4.3.5.3) to the ICF.

f) The ICF caches the received information as an ID association for the SUPI.

g) The UE acknowledges the receipt of the 5G-GUTI to the AMF using the REGISTRATION COMPLETE message. At this, the assignment of the 5G-GUTI is successful.

Scenario 2: 5G-GUTI in CONFIGURATION UPDATE COMMAND to the UE (Paging):

a) The AMF sends the Paging to the UE and the UE responds with a Service Request. The Service Request is expected to include the 5G-S-TMSI in clear.

b) At the conclusion of this Service Request handling, the AMF assigns a new 5G-GUTI to the UE and sends the same in the CONFIGURATION UPDATE COMMAND message.

c) The IEF present in the AMF detects the assignment of a new 5G-GUTI and sends the IEF Association Record message to the ICF that includes the 5G-GUTI, SUPI, Time-stamp = Tp, N-CGI, TAI, N-CGI Time (see clause 4.3.5.3) to the ICF.

f) The ICF caches the received information as ID association for the SUPI. The ICF begins the de-caching process for the previous ID association.

g) The UE acknowledges the receipt of the 5G-GUTI to the AMF using the CONFIGURATION UPDATE COMPLETE message. At this, the assignment of the new 5G-GUTI is successful.

Scenario 3: 5G-GUTI in CONFIGURATION UPDATE COMMAND to the UE (Service Request):

a) The AMF receives a Service Request from the UE. The Service Request is expected to include the 5G-S-TMSI in clear.

b) At the conclusion of this Service Request handling, the AMF assigns a new 5G-GUTI to the UE and sends the same in the CONFIGURATION UPDATE COMMAND message.

c) The IEF present in the AMF detects the assignment of a new 5G-GUTI and sends the IEF Association Record message to the ICF that includes the 5G-GUTI, SUPI, Time-stamp = Ty, N-CGI, TAI, N-CGI Time (see clause 4.3.5.3) to the ICF.

f) The ICF caches the received information as ID association for the SUPI. The ICF begins the de-caching process for the previous ID association.

g) The UE acknowledges the receipt of 5G-GUTI to the AMF using the CONFIGURATION UPDATE COMPLETE message. At this, the assignment of new 5G-GUTI is successful.

Scenario 4: 5G-GUTI in REGISTRATION ACCEPT to the UE (e.g. periodic registration requests):

a) The AMF may also receive a subsequent REGISTRATION REQUEST message from the UE. The REGISTRATION REQUEST may have the 5G-GUTI in clear.

b) At the conclusion of this REGISTRATION REQUEST handling, the AMF assigns a new 5G-GUTI to the UE and sends the same in the REGISTRATION ACCEPT message.

c) The IEF present in the AMF detects the assignment of a new 5G-GUTI and sends the IEF Association Record message to the ICF that includes the 5G-GUTI, SUPI, Time-stamp = Tw, N-CGI, TAI, N-CGI Time (see clause 4.3.5.3) to the ICF.

f) The ICF caches the received information as ID association for the SUPI. The ICF begins the de-caching process for the previous ID association.

g) The UE acknowledges the receipt of 5G-GUTI to the AMF using the REGISTRATION COMPLETE message. At this, the assignment of new 5G-GUTI is successful.

Scenario 5: GPSI is changed for the SUPI.

a) The AMF receives the AM Subscriber Data change notification from the UDM about the GPSI change.

b) The IEF present in the AMF detects the data change and sends the IEF Association Record message to the ICF that includes the 5G-GUTI, SUPI, GPSI, Time-stamp = Tx, N-CGI, TAI, N-CGI Time (see clause 4.3.5.3) to the ICF.

c) The ICF caches the received information as ID association for the SUPI. The ICF begins the de-caching process for the previous ID association.

4.3.6.2 De-caching scenarios

As explained in clause 4.3.3, there are two ways how an ID association in the ICF can be de-cached. The first method based on the caching of a new ID association is illustrated as a part of the caching scenarios in clause 4.3.6.1. This clause illustrates the second method where the IEF sends an IEF Disassociation Record message. Multiple possible scenarios in reference to this are depicted in figure 4.3.6.2-1 below.

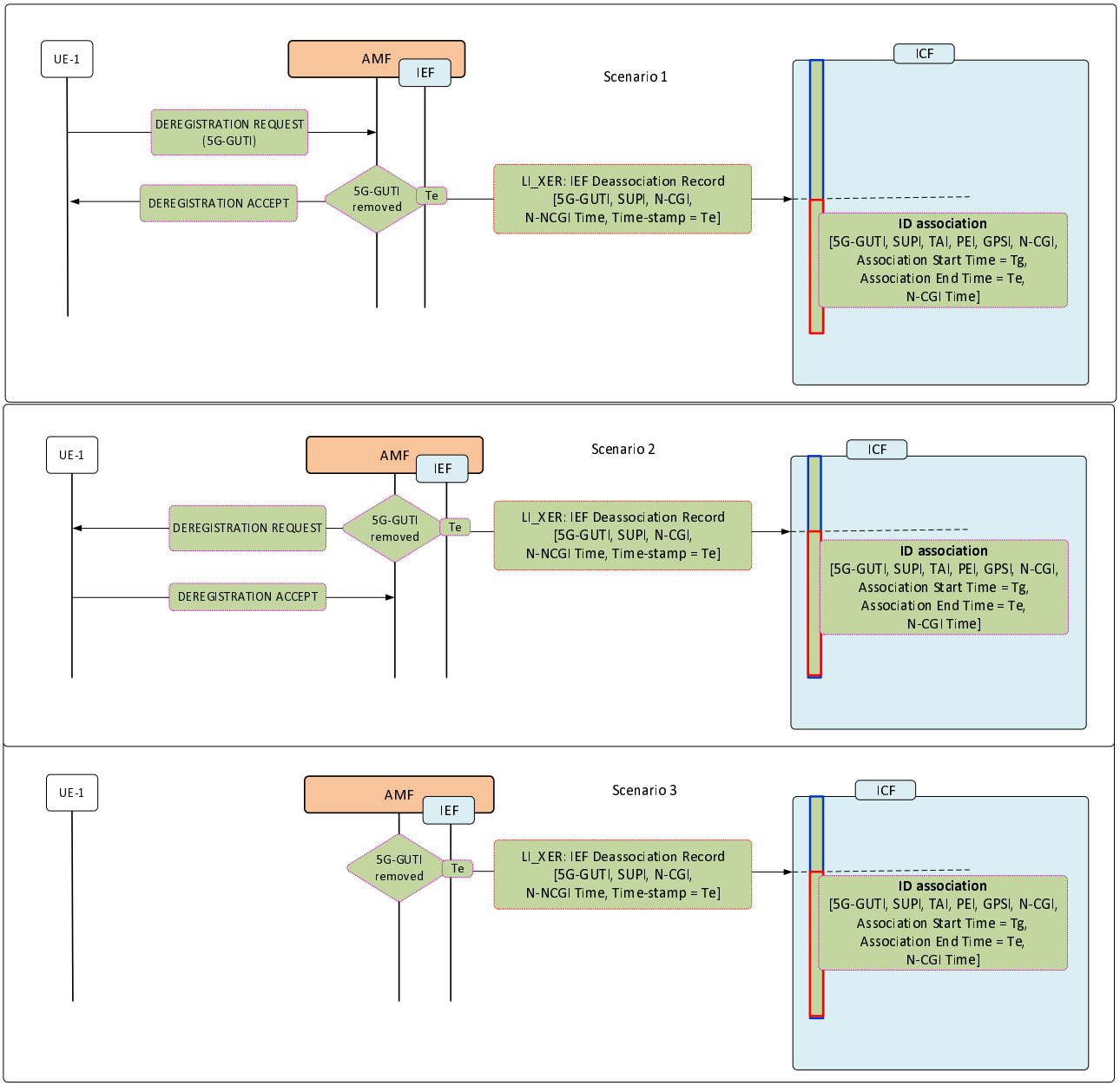


Figure 4.3.6.2-1: De-caching scenarios

Figure 4.3.6.2-1 shows 3 scenarios of ID association de-caching:

Scenario 1: UE deregisters from the network (e.g. graceful power down):

a) When the UE deregisters from the network, it sends a DEREGISTRATION REQUEST message to the AMF. The DEREGISTRATION REQUEST message contains the 5G-GUTI in clear.

b) The AMF disassociates the 5G-GUTI from the SUPI and sends a DEREGISTRATION ACCEPT message to the UE.

c) The IEF present in the AMF detects the removal of the 5G-GUTI for the SUPI and sends the IEF Disassociation Record message to the ICF that includes the 5G-GUTI, SUPI, Time-stamp = Te, N-CGI, N-CGI Time (see clause 4.3.5.3) to the ICF.

d) The ICF begins the de-caching process for the cached ID association.

Scenario 2: Network de-registers the UE:

a) The AMF when it determines to deregister a UE, removes the 5G-GUTI assigned for the SUPI and sends a DEREGISTRATION REQUEST to the UE.

b) The IEF present in the AMF detects the removal of the 5G-GUTI for the SUPI and sends the IEF Disassociation Record message to the ICF that includes the 5G-GUTI, SUPI, Time-stamp = Te, N-CGI, N-CGI Time (see clause 4.3.5.3) to the ICF.

c) The ICF begins the de-caching process for the cached ID association.

d) The UE acknowledges the DEREGISTRATION REQUEST with a DEREGISTRATION ACCEPT message. At this time, the deregistration of the UE is successful.

Scenario 3: Network removes the 5G-GUTI (e.g. battery removal):

a) When the network detects that the UE is no longer reachable, it may remove the associated 5G-GUTI for the SUPI.

b) The IEF present in the AMF detects the removal of the 5G-GUTI for the SUPI and sends the IEF Disassociation Record message to the ICF that includes the 5G-GUTI, SUPI, Time-stamp = Te, N-CGI, N-CGI Time (see clause 4.3.5.3) to the ICF.

c) The ICF begins the de-caching process for the cached ID association.

4.3.6.3 ID association and unsuccessful 5G-GUTI assignments

This is an error scenario where the 5G-GUTI assignment is unsuccessful. It can happen when the UE fails to receive a message that carries the new 5G-GUTI or the AMF fails to receive the acknowledgement from the UE. The two scenarios are illustrated below in figure 4.3.6.3-1.

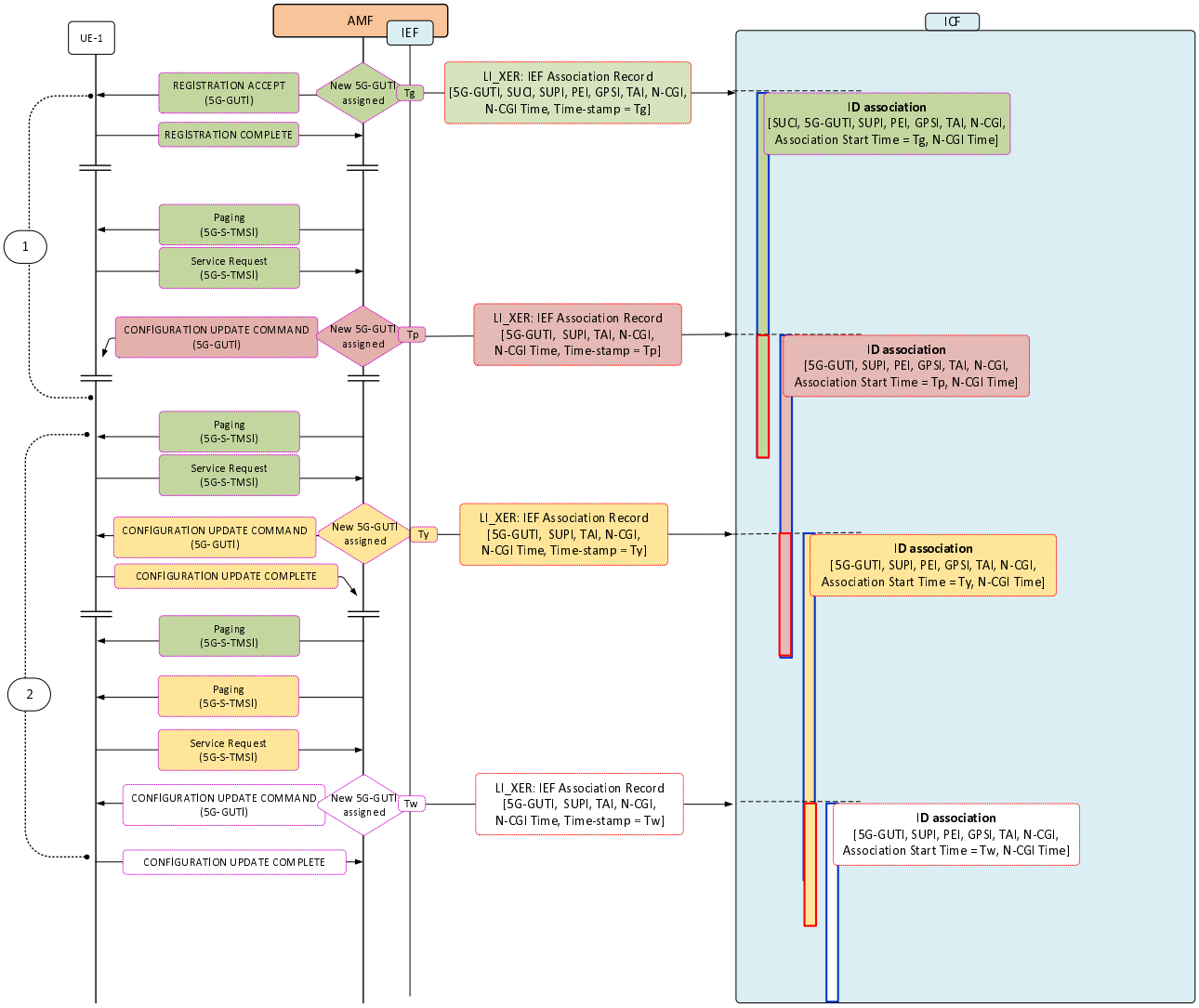


Figure 4.3.6.3-1: ID association and unsuccessful 5G-GUTI assignments

Figure 4.3.6.3-1 shows 2 scenarios of unsuccessful 5G-GUTI association and the ID Association Caching.

Scenario 1: CONFIGURATION UPDATE COMMAND does not reach the UE:

a) The AMF assigns a 5G-GUTI to the UE and sends the same to the UE in the REGISTRATION ACCEPT message.

b) The IEF present in the AMF detects the assignment of a new 5G-GUTI and sends the IEF Association Record message to the ICF that includes the 5G-GUTI, SUCI, SUPI, PEI, GPSI, Time-stamp = Tg, N-CGI, TAI, N-CGI Time to the ICF.

c) The ICF caches the received information as ID association for the SUPI.

d) The UE acknowledges the receipt of the 5G-GUTI to the AMF using the REGISTRATION COMPLETE message. At this, the assignment of the 5G-GUTI is successful.

e) The AMF sends the Paging to the UE and the UE responds with a Service Request. The Service Request is expected to include the 5G-S-TMSI in clear.

f) At the conclusion of this Service Request handling, the AMF assigns a new 5G-GUTI to the UE and sends the same in the CONFIGURATION UPDATE COMMAND message.

g) The IEF present in the AMF detects the assignment of a new 5G-GUTI and sends the IEF Association Record message to the ICF that includes the 5G-GUTI, SUPI, Time-stamp = Tp, N-CGI, TAI, N-CGI Time to the ICF.

h) ICF caches the received information as ID association for the SUPI. ICF begins the de-caching process for the previous ID association.

i) The CONFIGURATION UPDATE COMMAND does not reach the UE. The AMF, upon failing to receive the expected CONFIGURATION UPDATE COMPLETE message, keeps both the previous 5G-GUTI (first choice) and the new 5G-GUTI (second choice) as still being associated to the SUPI.

Within the ICF, the previous ID association will be deleted after the de-caching period. In this scenario, the active ID association in the ICF contains the second choice 5G-GUTI that the AMF is associating to the SUPI. However, that 5G-GUTI does not match to the 5G-GUTI (which is the first choice 5G-GUTI that AMF has) that the UE has.

Scenario 2: CONFIGURATION UPDATE COMPLETE does not reach the AMF:

a) The AMF sends the Paging to the UE using the first choice 5G-GUTI and the UE responds with a Service Request. The Service Request is expected to include the 5G-S-TMSI in clear.

b) At the conclusion of this Service Request handling, the AMF assigns a new 5G-GUTI to the UE and sends the same in the CONFIGURATION UPDATE COMMAND message.

c) The IEF present in the AMF detects the assignment of a new 5G-GUTI and sends the IEF Association Record message to the ICF that includes the 5G-GUTI, SUPI, Time-stamp = Ty, N-CGI, TAI, N-CGI Time to the ICF. The ICF transfers the PEI, GPSI from the previous record to the new record.

d) The ICF caches the received information as an ID association for the SUPI. The ICF begins the de-caching process for the previous ID association.

f) The UE acknowledges the receipt of 5G-GUTI to the AMF using the CONFIGURATION UPDATE COMPLETE message. However, CONFIGURATION UPDATE COMPLETE message does not reach the AMF. The AMF keeps both the previous 5G-GUTI (first choice) and the new 5G-GUTI (second choice) as still being associated to the SUPI.

Within the ICF, the previous ID association will be deleted after the de-caching period. In this scenario, the active ID association in the ICF does not contain the 5G-GUTI that the AMF is associating as the first choice 5G-GUTI to the SUPI. However, the active ID association in the ICF does correspond to the 5G-GUTI that has been accepted from the UE (which is the second choice 5G-GUTI in AMF).

Continuing the scenario 2:

h) The AMF sends the Paging to the UE using the second choice 5G-GUTI and the UE responds with a Service Request. The Service Request is expected to include the 5G-S-TMSI in clear.

i) At the conclusion of this Service Request handling, the AMF assigns a new 5G-GUTI to the UE and sends the same in the CONFIGURATION UPDATE COMMAND message.

j) The IEF present in the AMF detects the assignment of a new 5G-GUTI and sends the IEF Association Record message to the ICF that includes the 5G-GUTI, SUPI, Time-stamp = Tw, N-CGI, TAI, N-CGI Time to the ICF.

k) The ICF caches the received information as the ID association for the SUPI. The ICF begins the de-caching process for the previous ID association. The ICF transfers the PEI, GPSI from the previous record to the new record.

l) The UE acknowledges the receipt of the 5G-GUTI to the AMF using the CONFIGURATION UPDATE COMPLETE message. At this, the assignment of 5G-GUTI is successful.

4.4 Retrieval principles

4.4.1 Overview

The second part of ID Association Caching is the retrieval of cached ID associations by the LEAs for a specific UE. The retrieval process is handled using the request and response procedures between the LEAs and the CSP.

A LEA can submit six types of requests to the CSP:

1. Request for the SUPI associated with a SUCI.
2. Request for the SUPI associated with a 5G-S-TMSI.
3. Request for the SUPI associated with a 5G-GUTI.
4. Request for the 5G-GUTI associated with a SUPI.
5. Request for ongoing updates of new 5G-GUTI assigned to a SUPI.
6. Request to stop the above ongoing updates.

The first three are also referred to the retrieval of a permanent ID from the non-permanent ID and the next two are referred as a retrieval of non-permanent ID from a permanent ID.

As shown in figure 4.1-1, the LEA submits a request over the LI_HIQR interface to the IQF, which in turn forwards that request to the ICF over the LI_XQR interface. The response to such a request involves the ICF retrieving the cached ID association and returning the same to the LEA via the IQF.

The ICF retrieves the cached ID association known at the time it receives the request unless the LEA specifies a particular time of interest to retrieve the ID association. The specific time which may be included in some of the requests is referred to as Observed Time. In the list shown above, the LEA includes the Observed Time in the first 4 of the requests.

The ICF within the CSP uses the Observed Time after applying a short time-window to overcome any timing issues between the LEA clock and the CSP clock to retrieve the cached ID association. The Observed Time is compared against the Association Start Time stored in the cached ID association.

4.4.2 LI_HIQR

The LI_HIQR is an interface that carries requests and responses between the LEA and the IQF.

On the LI_HIQR interface, the following names are used to for the messages exchanged between the LEA and the IQF:

- LI_HIQR Request (LEA to IQF).
- LI_HIQR Response (IQF to LEA).
- LI_HIQR Updates (IQF to LEA).

The LI_HIQR Request has to always carry the Observed Time field when the ID included in the message is the 5G-S-TMSI or the 5G-GUTI or the SUPI except for the case of ongoing updates. The TS 33.128 [4] defines the details of the LI_HIQR Request, LI_HIQR Response and LI_HIQR Updates. The same is summarized in the table below.

Table 4.4.2-1: LI_HIQR Request possibilities

Message Name	Request Details			
	Request Type	Observed Time	Request Values	
LI_HIQR Request	Identity Association	Present	SUCI	N-CGI, TAC
		Present	5G-S-TMSI	N-CGI, TAC
		Present	5G-GUTI	N-CGI, TAC
		Present	SUPI	N/A
	Ongoing Identity Association	Not present	SUPI	N/A

In table 4.4.2-1, the Request Type of Identity Association with Request Values of SUCI, 5G-S-TMSI, and 5G-GUTI may also be viewed as a request for a non-permanent ID to a permanent ID. The Request Type of Identity Association with Request Values of SUPI and the Ongoing Identity Association may also be viewed as a request for a permanent ID to a non-permanent ID.

Table 4.4.2-2: LI_HIQR Response/Updates possibilities

Message Name	Identity Response Details						
	Identity Association Record (one or more instance, NOTE 1)						
	SUPI	5G-GUTI	SUCI	PEI, GPSI	Association		List of TA
					Start Time	End Time	
LI_HIQR Response	Present	Present	If found	If found	Present	When applicable	If available
LI_HIQR Updates	Present	Present	If found	If found	Present	When applicable	If available

NOTE 1: LI_HIQR Updates will have at most one Identity Association Record. The LI_HIQR Response may have more than one Identity Association Records.

In table 4.4.2-2, the Association Start Time included in the LI_HIQR Response and the LI_HIQR Updates reflects the time at which the 5G-GUTI is assigned to the SUPI in the AMF. The Association End Time is included in the LI_HIQR Response and the LI_HIQR Updates, when the ID association reported in the Identity Association Record is marked for deletion after the de-caching period within the ICF.

NOTE 2: The Association Start Time for the SUCI to SUPI retrieval case is also the time at which the 5G-GUTI is assigned in the AMF.

On the LI_HIQR interface, every message is acknowledged by the receiving function. For example: the LI_HIQR Request will have an acknowledgement. Likewise, the LI_HIQR Response will have an acknowledgement. In other words, the LI_HIQR Response is not really an acknowledgement to a LI_HIQR Request at a protocol level.

4.4.3 LI_XQR

The LI_XQR is the interface that carries requests and responses between the IQF and the ICF.

On the LI_XQR interface, the following names are used for the messages exchanged between the IQF and the ICF:

- Identity Association Request (IQF to ICF).
- Identity Association Response (ICF to IQF).
- Activate Association Update (IQF to ICF).
- Activate Association Update Acknowledgement (ICF to IQF).
- Identity Association Update (ICF to IQF).
- Identity Association Update Acknowledgement (IQF to ICF).
- Deactivate Association Update (IQF to ICF).
- Deactivate Association Update Acknowledgement (ICF to IQF).

TS 33.128 [4] defines the details of the above listed messages.

Table 4.4.3-1 provides a mapped LI_HIQR message for each of the above:

Table 4.4.3-1: Mapping between LI_HIQR and LI_XQR

LI_XQR		LI_HIQR		
Message Name	Direction	Message Name	Request Type	Direction
Identity Association Request	IQF to ICF	LI_HIQR Request	Identity Association	LEA to IQF
Identity Association Response	ICF to IQF	LI_HIQR Response	N/A	IQF to LEA
Activate Association Update	IQF to ICF	LI_HIQR Request	Ongoing Identity Association	LEA to IQF
Activate Association Update Acknowledgement	ICF to IQF	LI_HIQR Response	N/A	IQF to LEA
Identity Association Update	ICF to IQF	LI_HIQR Updates	N/A	IQF to LEA
Identity Association Update Acknowledgement	IQF to ICF	N/A	N/A	N/A
Deactivate Association Update	IQF to ICF	LI_HIQR Request	Ongoing Identity Association	LEA to IQF
Deactivate Association Update Acknowledgement	ICF to IQF	N/A	N/A	N/A

As shown in table 4.4.3-1, the Identity Association Update Acknowledgement is an acknowledgement from the IQF to the ICF for the receipt of Identity Association Update Message. This is not seen on the LI_HIQR interface. Also, the Deactivate Association Update Acknowledgement message, which is a response from ICF to IQF for indicating successfully stopping on the further reporting of new 5G-GUTI assignments, is not relayed to the LEA over LI_HIQR.

Table 4.4.3-2 provides the details of LI_XQR Identity Association Request possibilities.

Table 4.4.3-2: LI_XQR Identity Association Request possibilities

Message Name	Request Details			
	Request Type	Observed Time	Request Values	
Identity Association Request	Identity Association	Present	SUCI	N-CGI, TAC
		Present	5G-S-TMSI	N-CGI, TAC
		Present	5G-GUTI	N-CGI, TAC
		Present	SUPI	N/A

Table 4.4.3-3 shows the LI_XQR Ongoing Identity Association possibilities.

Table 4.4.3-3: LI_XQR Ongoing Identity Association possibilities

Message Name	SUPI	LD Task Object (LI_HIQR)
Activate Association Update	Present	Awaiting Disclosure
Deactivate Association Update	Present	Disclosed

The LD Task Object shown in the table 4.4.3-3 is not part of the LI_XQR messages. The IQF uses that information received in the LI_HIQR Request to generate the two messages over LI_XQR when the Request Type in the LI_HIQR Request is Ongoing Identity Association.

Table 4.4.3-4 provides the details of LI_XQR Response possibilities.

Table 4.4.3-4: LI_XQR Response possibilities

Message Name	Identity Response Details						
	Identity Association Record (one or more instance)						
	SUPI	5G-GUTI	SUCI	PEI, GPSI	Association		List of TA
					Start Time	End Time	

Identity Association Response	Present	Present	If found	If found	Present	When applicable	If available
Activate Association Update Acknowledgement	Present	Present	If found	If found	Present	N/A	If available
Identity Association Update	Present	Present	If found	If found	Present	When applicable	If available

The Identity Association Update Acknowledgement and the Deactivate Association Update Acknowledgement message are not seen on the LI_HIQR interface and therefore, the details of the same are not shown in table 4.4.3-4.

The Request Type is not included in the Activate Association Update or Deactivate Association Update messages. The LD Task Object Desired Status will have "Awaiting Disclosure" in all requests except for the Deactivate Association Update, in which case, it will be "Disclosed".

4.4.4 Illustration of retrieval principles

4.4.4.1 General

The LEA may send an LI_HIQR Request for the ID retrieval with an Observed Time or without an Observed Time. As described in clause 4.4.2 and table 4.4.2-1, all LI_HIQR Request messages except where the Request Type is Ongoing Identity Request will have an Observed Time.

4.4.4.2 Retrieval with Observed Time

When the Observed Time is in the LI_XQR: Identity Association Request, the ICF uses the Observed Time to search for cached ID associations. An ID association will be part of the response if the Association Start Time happens to be before the Observed Time, even if the caching at the ICF happened after the Observed Time. This is illustrated in figure 4.4.4.2-1 below.

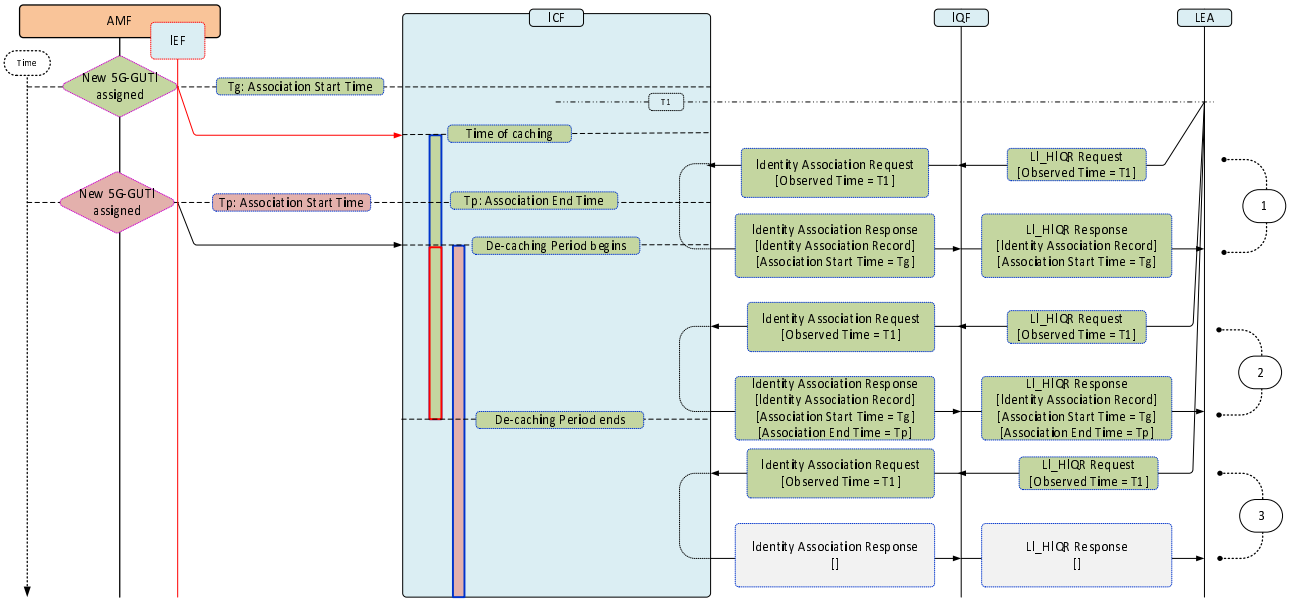


Figure 4.4.4.2-1: Retrieval with Observed Time

NOTE: The focus of the above figure is the retrieval that includes the LEA, the IQF and the ICF. That is why the arrows between the IEF and the ICF do not show the message names.

Figure 4.4.4.2-1 illustrates three scenarios:

Scenario 1:

- The request is received while the 5G-GUTI with an Association Start Time of Tg (which occurs before the Observed Time T1) is still active as per the cached ID association. Accordingly, the response will have that Identity Association Record with the Association Start Time of Tg.

Scenario 2:

- The request is received while the ID association of the 5G-GUTI with an Association Start Time of Tg (which occurs before the Observed Time T1) is in a de-caching phase. The Association Start Time Tp in the ID association of the next 5G-GUTI is after the Observed Time T1. Therefore, the Identity Association Response will have one Identity Association Record (of the ID association in a de-caching phase) showing the Association Start Time of Tg and Association End Time of Tp.

Scenario 3:

- The request is received while the 5G-GUTI with an Association Start Time of Tp (which occurs after the Observed Time T1) is active as per the cached ID association. However, since Tp is after the T1, the response will not have any Identity Association Records.

4.4.4.3 Retrieval without Observed Time

When the Observed Time is not present in the LI_XQR: Identity Association Request, the ICF retrieves the current active ID association and returns the same as the Identity Association Record within the Identity Association Response to the IQF. This is illustrated in figure 4.4.4.3-1 below.

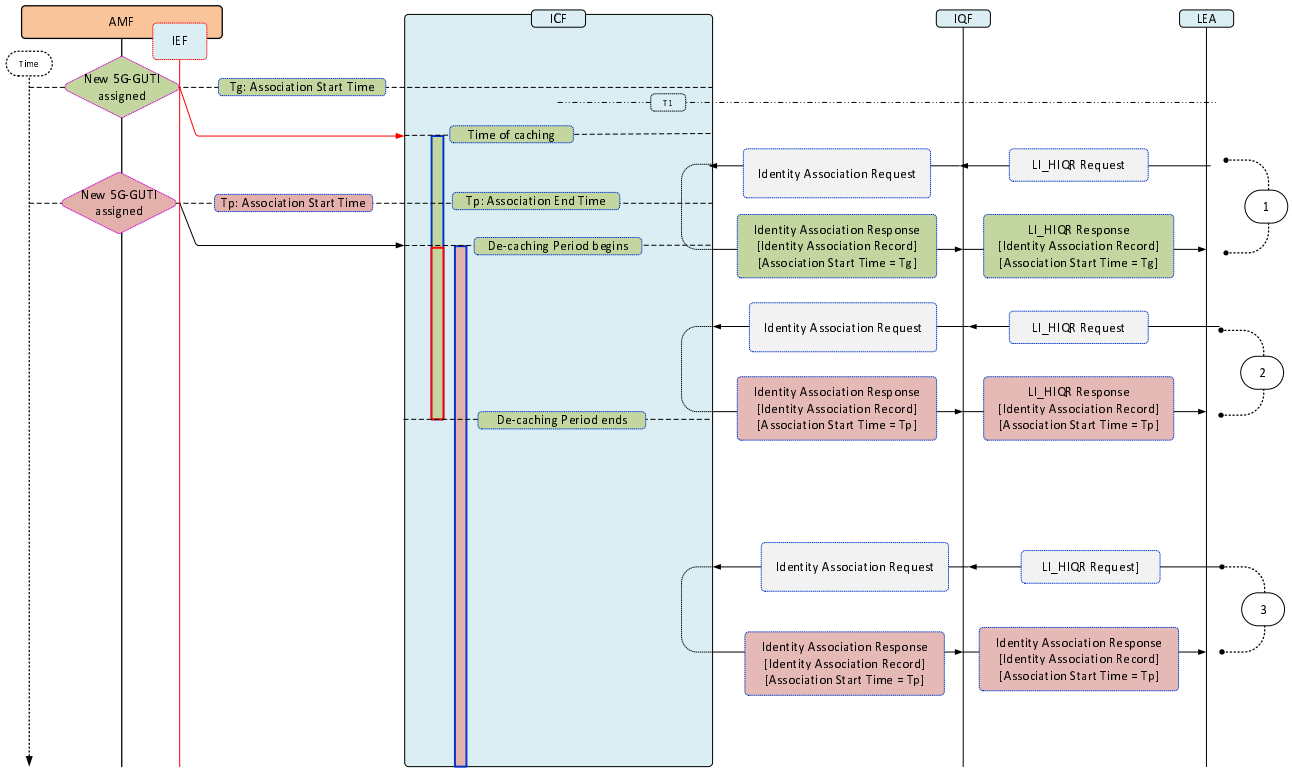


Figure 4.4.4.3-1: Retrieval without the Observed Time

NOTE: The focus of the above figure is on the ID association caching and de-caching. That is why the arrows between the IEF and ICF do not show the message names.

Figure 4.4.4.3-1 illustrates three scenarios:

Scenario 1:

- The request is received while the 5G-GUTI with an Association Start Time of Tg is active as per the cached ID association. Accordingly, the response will include that Identity Association Record which has the Association Start Time of Tg.

Scenario 2:

- The request is received while the ID association of the 5G-GUTI with an Association Start Time of Tg is in a de-caching phase. The next 5G-GUTI with an Association Start Time Tp is active as per the cached ID association record. Therefore, the response will include that Identity Association Record which has the Association Start Time of Tp.

Scenario 3:

- The request is received while the ID association of the 5G-GUTI with an Association Start Time of Tg is deleted from the ICF. The 5G-GUTI with an Association Start Time Tp is active as per the cached ID association record. Therefore, the response will include that Identity Association Record which has the Association Start Time of Tp.

4.4.5 Illustration of time-window principles

4.4.5.1 Overview

The concept of a short time-window is used within the ICF while searching for the ID associations for the LEA requests that contain the Observed Time. The reason for using such a time-window is to overcome any possible differences between the LEA clock and the CSP clock.

NOTE 1: The use of such a short time-window may only be needed for permanent ID to non-permanent ID retrieval case.

NOTE 2: The illustrations assume that the message transfers and the caching happen instantaneously.

4.4.5.2 No short time-window

4.4.5.2.1 Scenario 1 – two clocks are synched

Figure 4.4.5.2.1-1 illustrates a scenario where the two clocks (CSP and LEA) are synchronized, and no short time-window is maintained within the ICF.

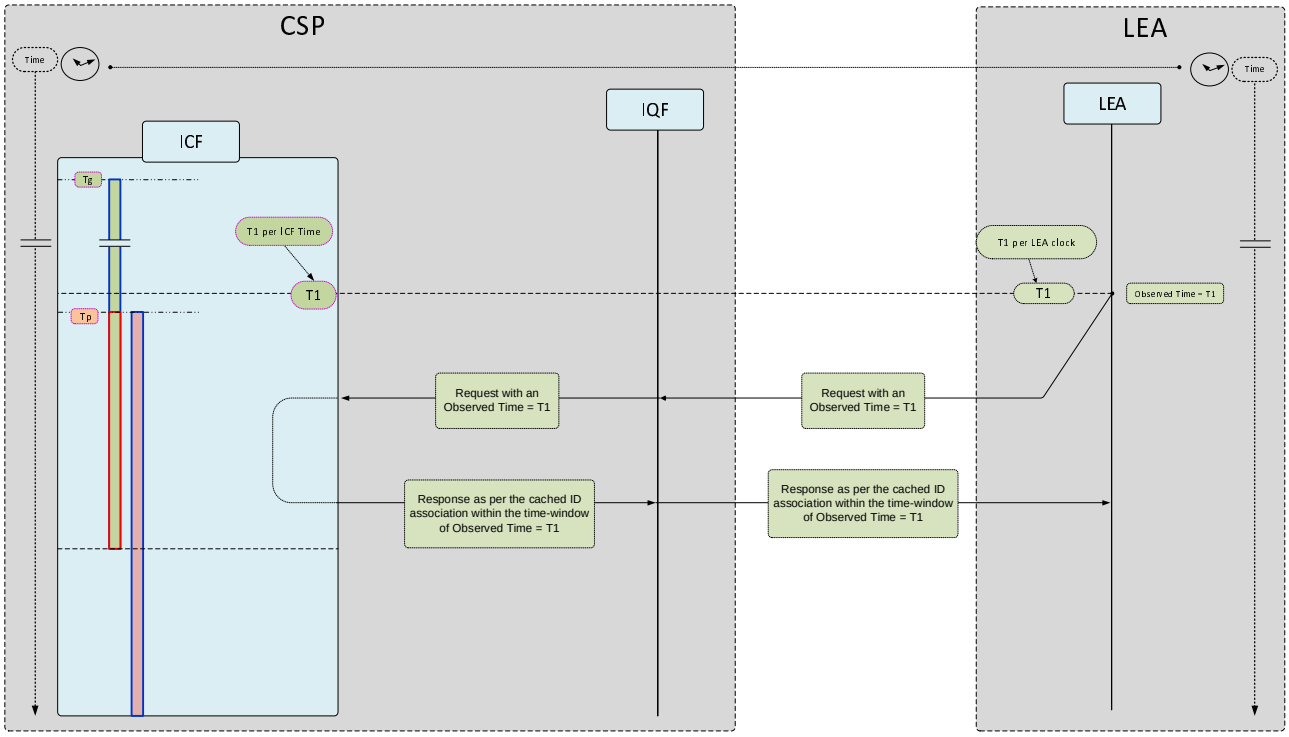


Figure 4.4.5.2.1-1: Scenario 1 – CSP clock is synchronized with LEA clock – no short time-window in ICF

In the illustration shown in figure 4.4.5.2.1-1, one ID association is in the ICF (cached at Tg) at the time T1. The response will, therefore, have one Identity Association Record.

4.4.5.2.2 Scenario 2 – CSP clock lags LEA clock

Figure 4.4.5.2.2-1 illustrates a scenario where the CSP clock lags the LEA clock, but no short time-window is maintained within the ICF.

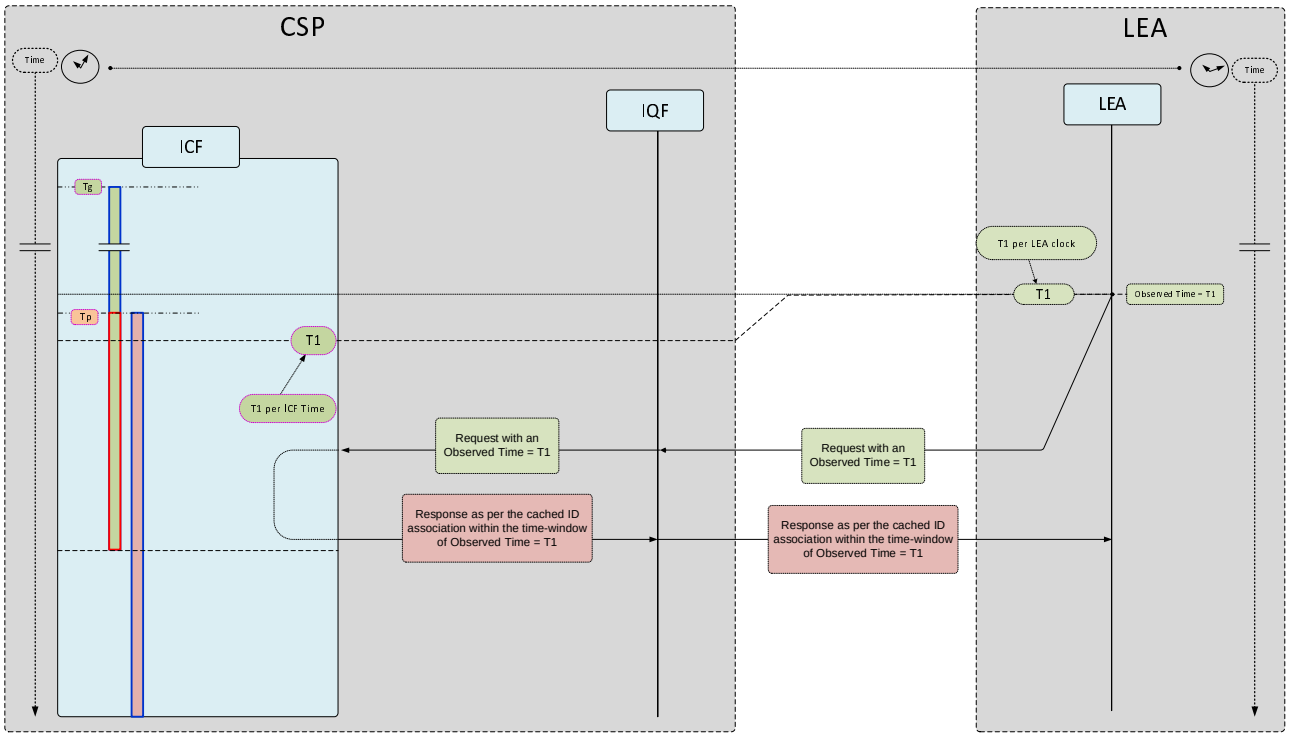


Figure 4.4.5.2.2-1: Scenario 2 – CSP clock lags LEA clock – no short time-window in ICF

In the illustration shown in figure 4.4.5.2.2-1, since the CSP clock lags the LEA clock, the time T1 in the ICF is yet to happen in reference to the Observed Time T1 per LEA clock. At time T1 (per the CSP clock), two ID associations are in the ICF (cached at Tg and Tp), but only the one with the Association Start Time Tp is active. The response will, therefore, have one Identity Association Record.

4.4.5.2.3 Scenario 3 – LEA clock lags CSP clock

Figure 4.4.5.2.3-1 illustrates a scenario where the LEA clock lags the CSP clock, but no short time-window is maintained within the ICF.

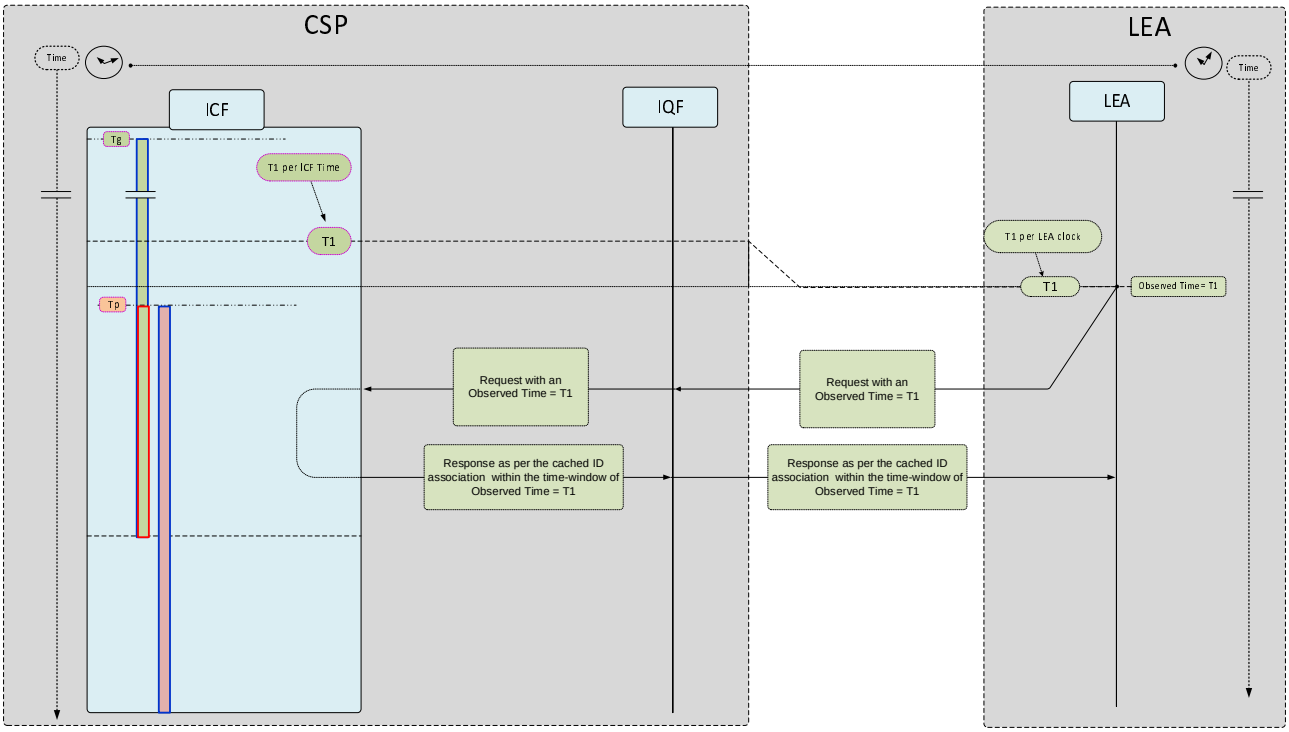


Figure 4.4.5.2.3-1: Scenario 3 – LEA clock lags CSP clock – no short time-window in ICF

In the illustration shown in figure 4.4.5.2.3-1, since the LEA clock lags the CSP clock, the time T1 in ICF has already happened in reference to the Observed Time T1 per LEA clock. At time T1 (per the CSP clock), one ID association is in the ICF (cached at Tg). The response will, therefore, have one Identity Association Record. The net result of this one is the same as in scenario 1.

4.4.5.3 Short time-window

4.4.5.3.1 Scenario 1 – two clocks are synched

The figure 4.4.5.3.1-1 illustrates a scenario where the two clocks (CSP and LEA) are synchronized and the ICF has implemented the short time-window concept.

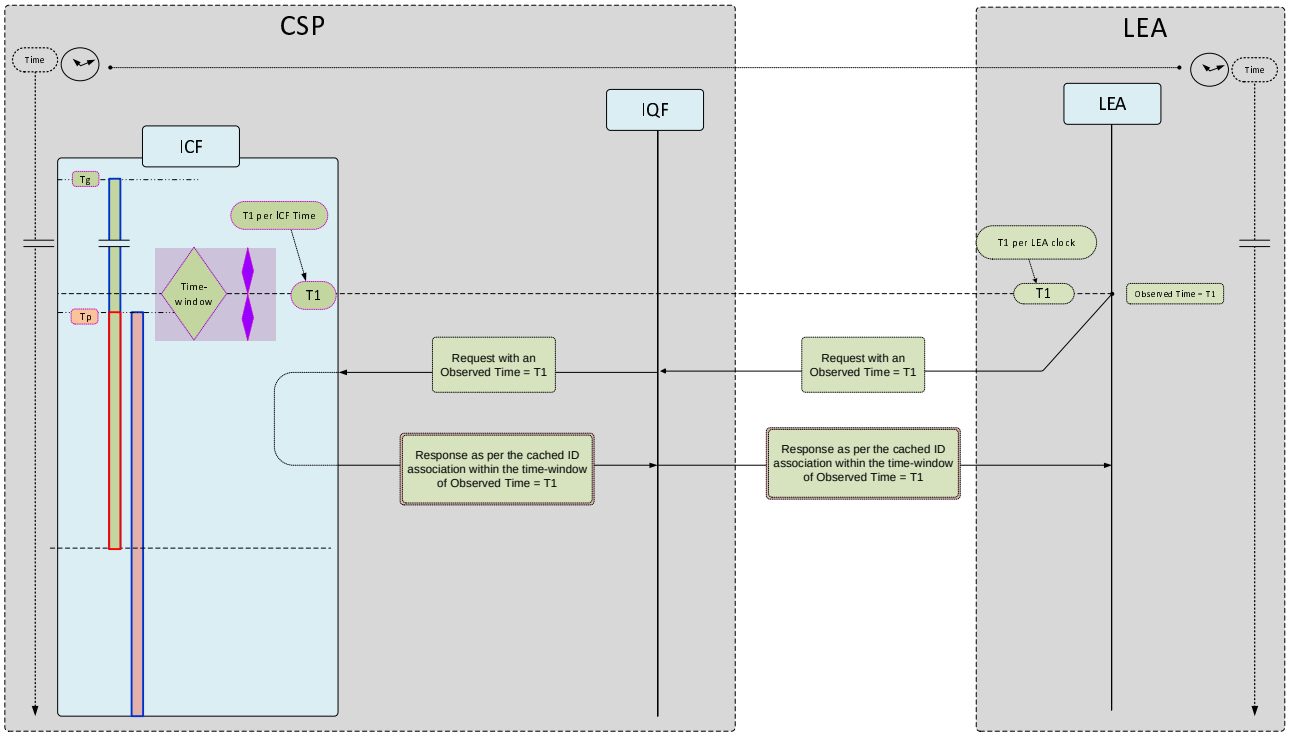


Figure 4.4.5.3.1-1: Scenario 1 – CSP clock is synchronized with LEA clock – short time-window in ICF

In the illustration shown in figure 4.4.5.3.1-1, one ID association is in the ICF (cached at Tg) at the time T1. However, when the ICF is scanned with a short time-window, the ID association (cached at Tp) can also be viewed as the ID association at time T1. The response will, therefore, have two Identity Association Records.

4.4.5.3.2 Scenario 2 – CSP clock lags LEA clock

Figure 4.4.5.3.2-1 illustrates a scenario where the CSP clock lags the LEA clock and ICF has implemented the short time-window concept.

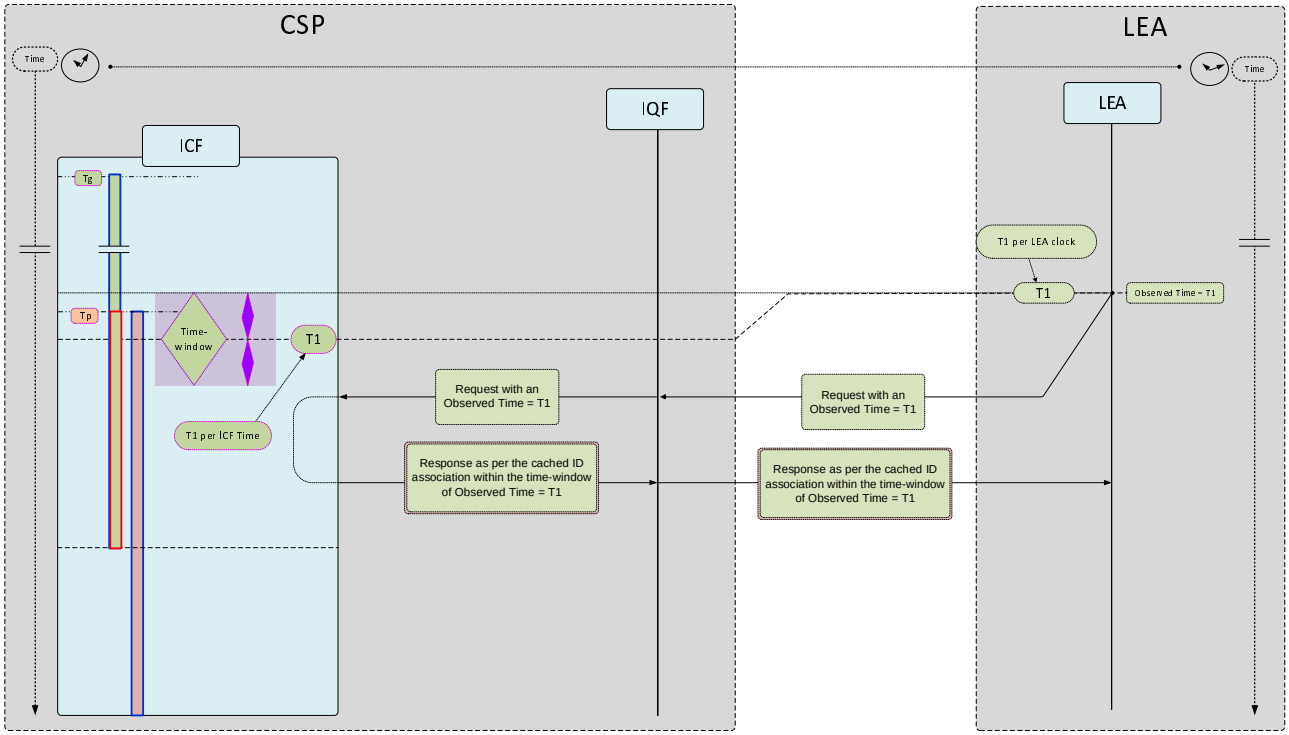


Figure 4.4.5.3.2-1: Scenario 2 – CSP clock lags LEA clock – short time-window in ICF

In the illustration shown in figure 4.4.5.3.2-1, since the CSP clock lags the LEA clock, the time T1 in the ICF is yet to happen in reference to the Observed Time T1 per LEA clock. At time T1 (per the CSP clock), two ID associations are in the ICF (cached at Tg and Tp) and both can be considered active when the Time-Window is applied. The response will, therefore, have two Identity Association Records.

4.4.5.3.3 Scenario 3 – LEA clock lags CSP clock

Figure 4.4.5.3.3-1 illustrates a scenario where the LEA clock lags the CSP clock and ICF has implemented the short time-window concept.

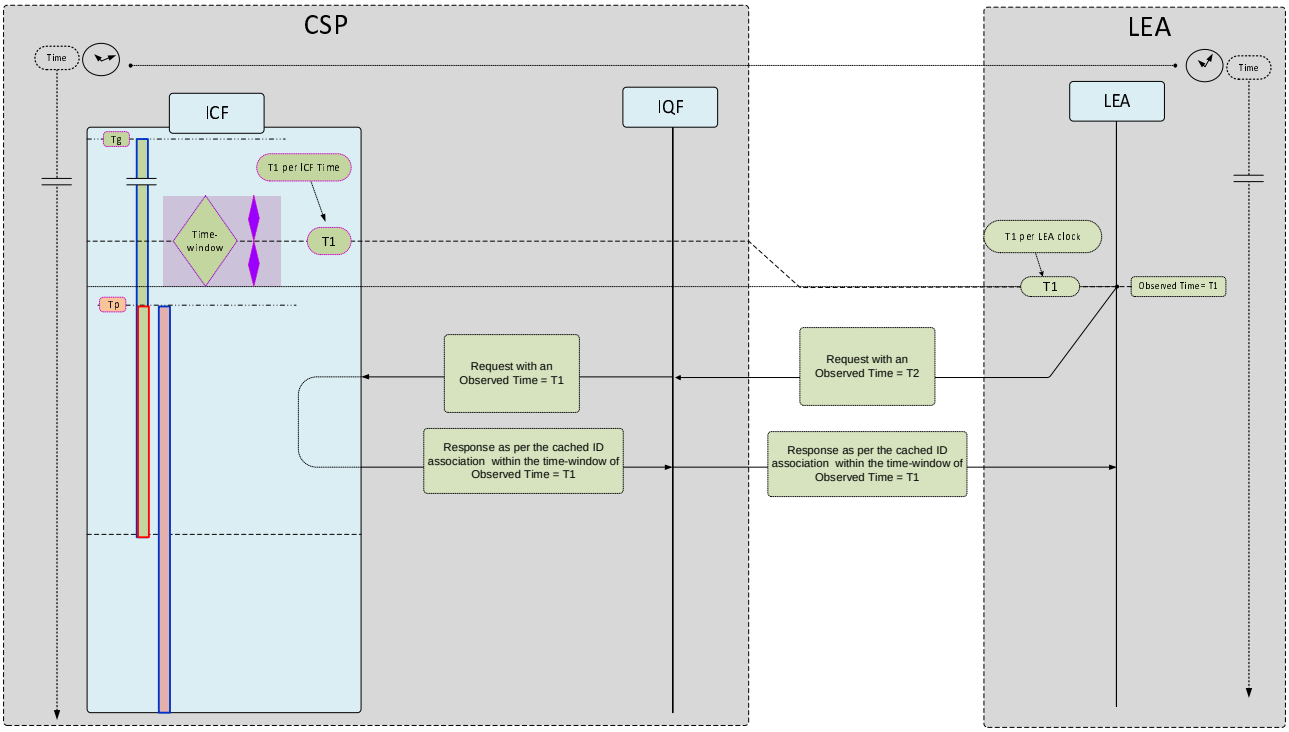


Figure 4.4.5.3.3-1: Scenario 3 – LEA clock lags CSP clock – short time-window in ICF

In the illustration shown in figure 4.4.5.3.3-1, since the LEA clock lags the CSP clock, the time T1 in ICF has already happened in reference to the Observed Time T1 per LEA clock. At time T1 (per the CSP clock), one ID association is in the ICF (cached at Tg) and the same will be the result even if the ICF is scanned with a short time-window. The response will, therefore, have one Identity Association Record.

4.4.5.3.4 Scenario 4 - no previous ID association in the ICF (LEA clock lags)

If there is no prior ID association cached in the ICF, then the scenario 3 as illustrated in figure 4.4.5.3.3-1 may not find any ID association in the ICF. In that case, the response will have zero Identity Association Records even if the ICF maintains a short time-window.

This is illustrated in figure 4.4.5.3.4-1 below.

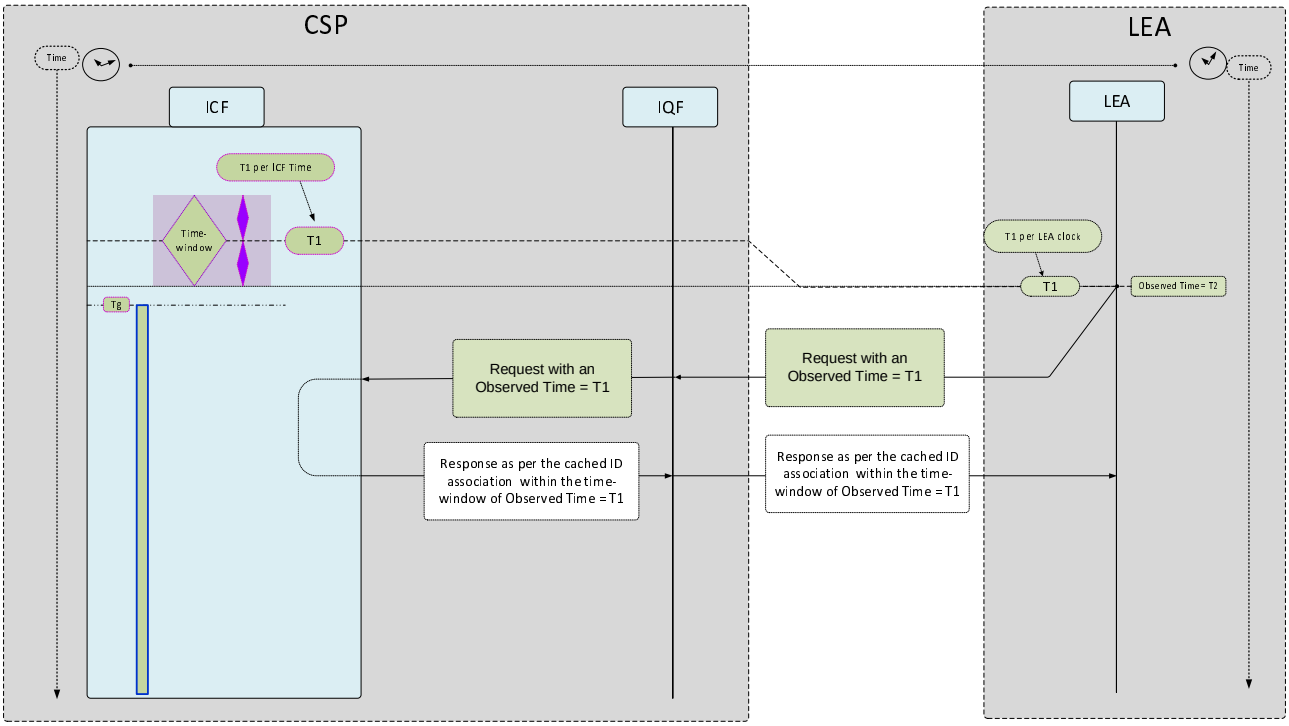


Figure 4.4.5.3.4-1: Scenario 4 – LEA clock lags CSP clock – no prior ID association in the ICF

As shown, at the time T1 even with a short time-window, no ID association is in the ICF. Therefore, even though the ICF may have an ID association when the LI_XQR Request is received, the response which should be based on the Observed Time T1, will have zero Identity Association Records.

4.4.5.3.5 Scenario 5 - no previous ID association in the ICF (CSP clock lags)

If there is no prior ID association cached in the ICF, then the scenario 4 illustrated in figure 4.4.5.3.4-1 may still find an ID association in the ICF when the CSP clock lags the LEA clock as shown in figure 4.4.5.3.5-1 below.

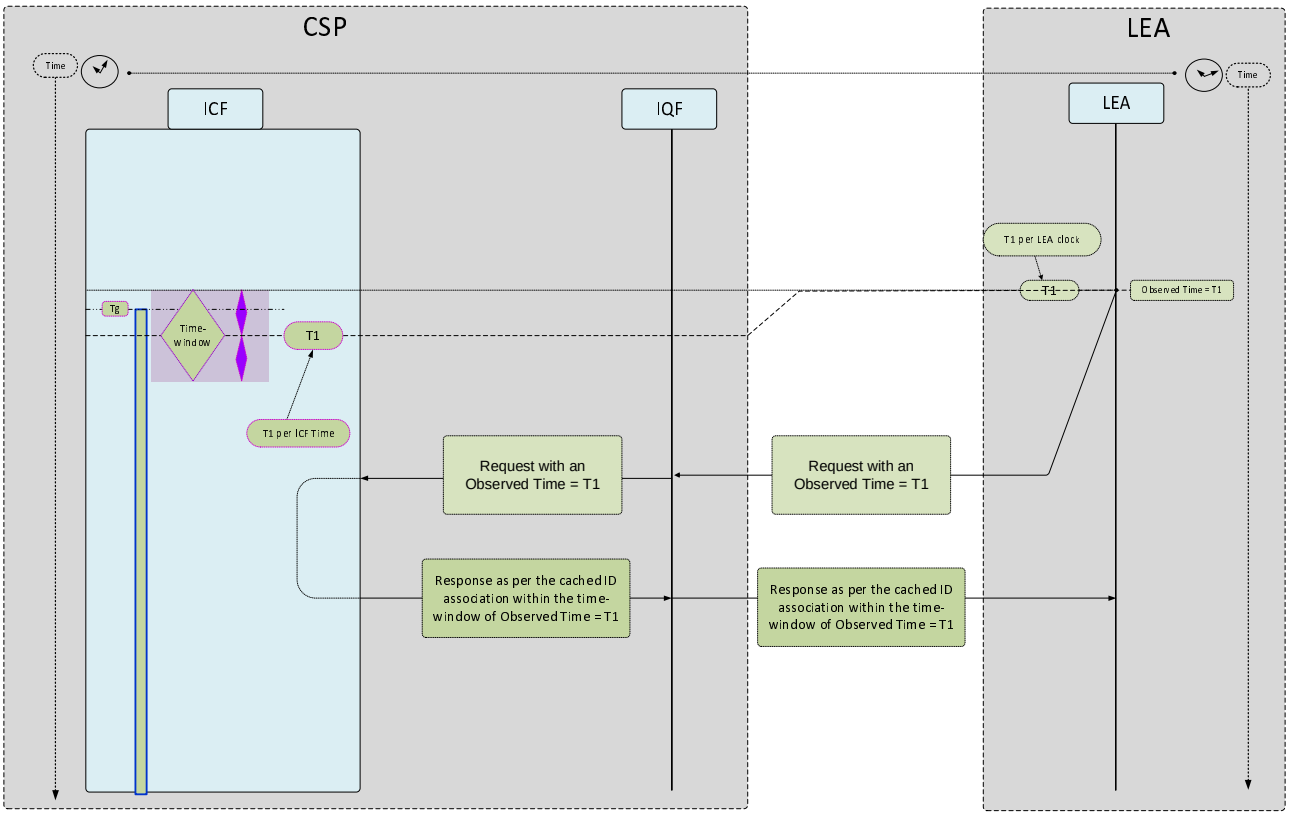


Figure 4.4.5.3.5-1: Scenario 5 – CSP clock lags LEA clock – no prior ID association in the ICF

As shown, at the time T1 (as per the ICF time), an ID association is in the ICF and therefore, the response will have an Identity Association Record.

4.4.5.3.6 Scenario 6 – longer time-window is a possible implementation

Since the ICF does not know the time-difference between the two clocks, the value of the short time-window is to be determined by the CSP. It can be longer than the actual time-difference, if measured between the CSP clock and the LEA clock. The impact of this for the scenario 4 is illustrated below in figure 4.4.5.3.6-1 below.

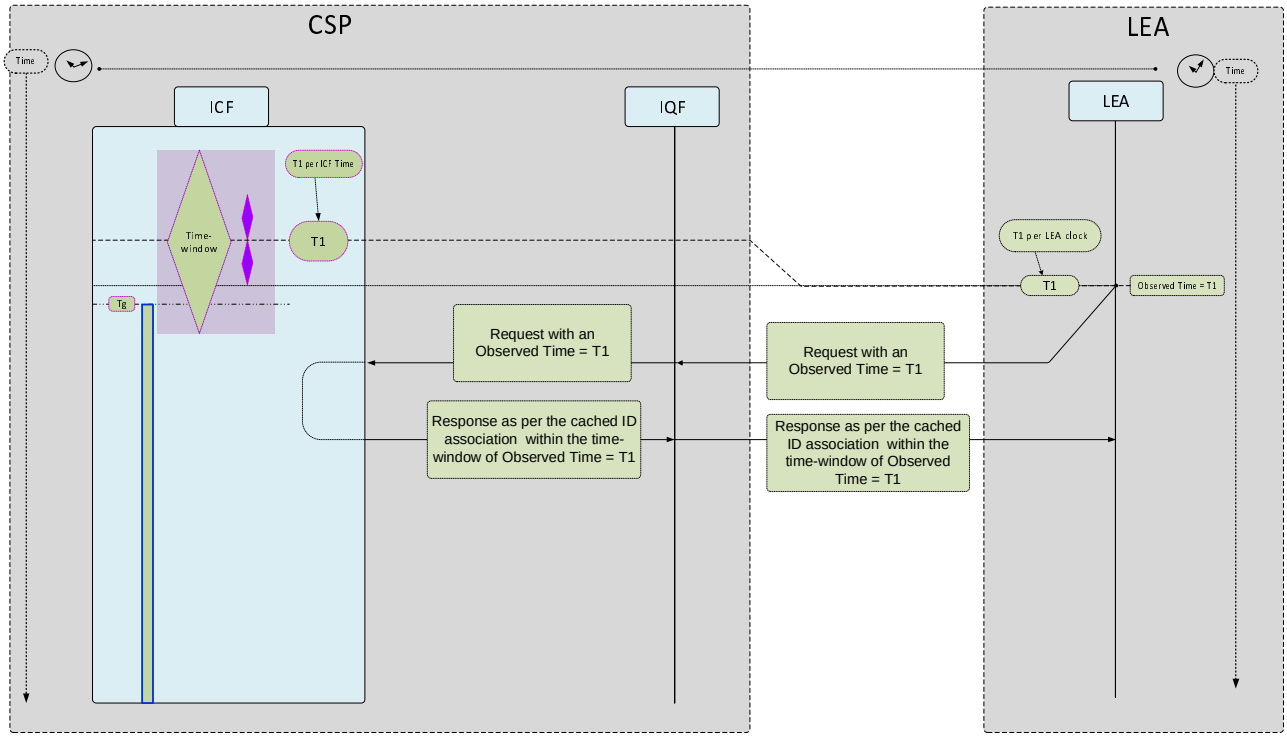


Figure 4.4.5.3.6-1: Scenario 6 – LEA clock lags CSP clock - longer time-window

As shown, even though no ID association is cached in the ICF at the time T1 per ICF time, with the time-window concept, the ICF may be able to find the ID association record. In this case, the response will have one ID Association Record.

NOTE: Even though a longer time-window at times may help, it is possible that such a long time-window may also result in unwanted results.

4.5 Flow diagrams

4.5.1 General

The purpose of this clause is to show end-to-end flow diagrams depicting the scenarios of ID Association Caching with the scope of showing various caching scenarios combined with the retrieval scenarios.

For the simplicity of the drawings, all the flow-diagrams assume that the LEA clock and the CSP clock are aligned. The ICF may still have the short time window, described and illustrated in clause 4.4.

4.5.2 SUCI to SUPI retrieval

4.5.2.1 General

As illustrated in the caching principles, the ID association is cached in the ICF whenever a 5G-GUTI is assigned to the SUPI.

The association of a SUPI to a SUCI may happen in the AMF before a 5G-GUTI is assigned. Therefore, it is possible that the Observed Time provided by the LEA Request can be before the Association Start Time of the first cached ID association. In that case the response will not have any ID Association Records.

The LEA may submit a request with a later time, for example, the time of the 5G-GUTI assignment observation. In that case, the Identity Association Response may carry the Identity Association Records.

4.5.2.2 Illustration 1

Figure 4.5.2.2-1 illustrates the two scenarios of the SUCI to SUPI retrieval handling case with the Observed Time happenings before the Association Start Time of the first cached ID association.

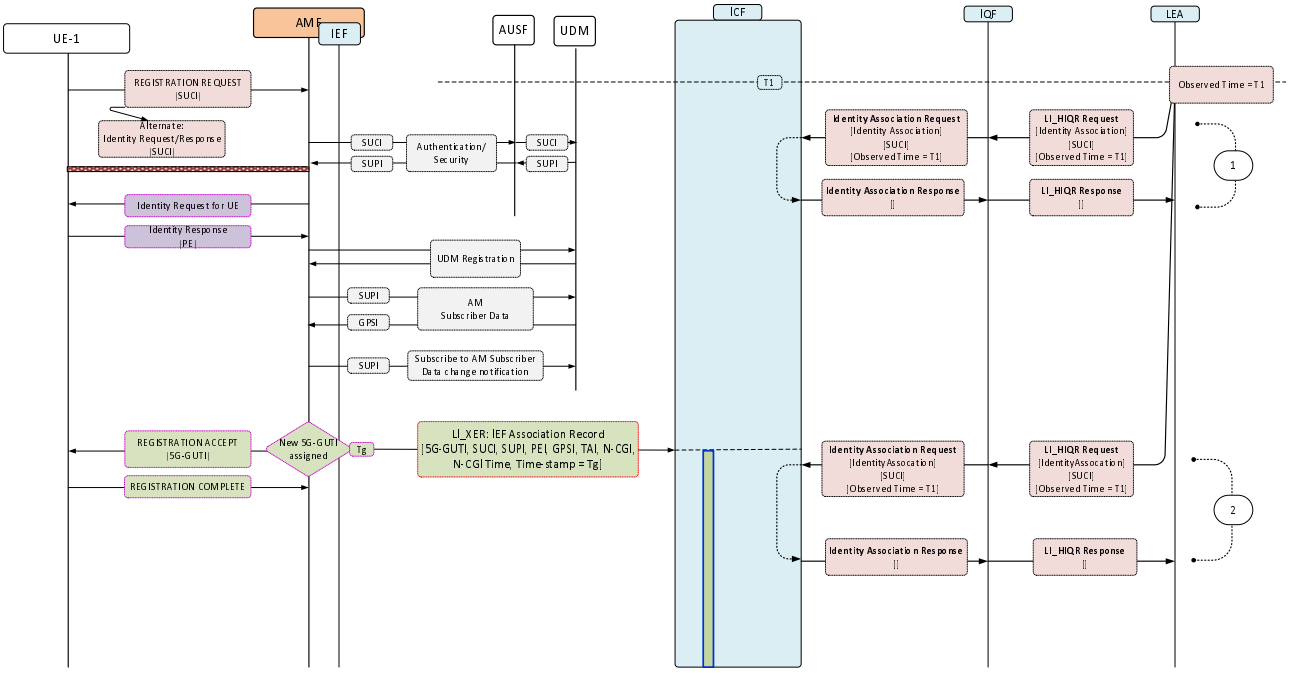


Figure 4.5.2.2-1: SUCI to SUPI retrieval – illustration 1

Figure 4.5.2.2-1 illustrates two scenarios:

Scenario 1:

- The Identity Association Request is received at the ICF before the ICF had the ID association caching for the SUCI. Accordingly, the Identity Association Response will have no Identity Association Records.

Scenario 2:

- The Identity Association Request is received at the ICF while the ICF has the ID association caching for the SUCI. However, the Observed Time is before the Association Start Time Tg in the cached ID association. Accordingly, the Identity Association Response will have no Identity Association Records.

4.5.2.3 Illustration 2

Figure 4.5.2.3-1 illustrates the two scenarios of SUCI to SUPI retrieval handling case with the Observed Time happenings after the Association Start Time of the first cached ID association.

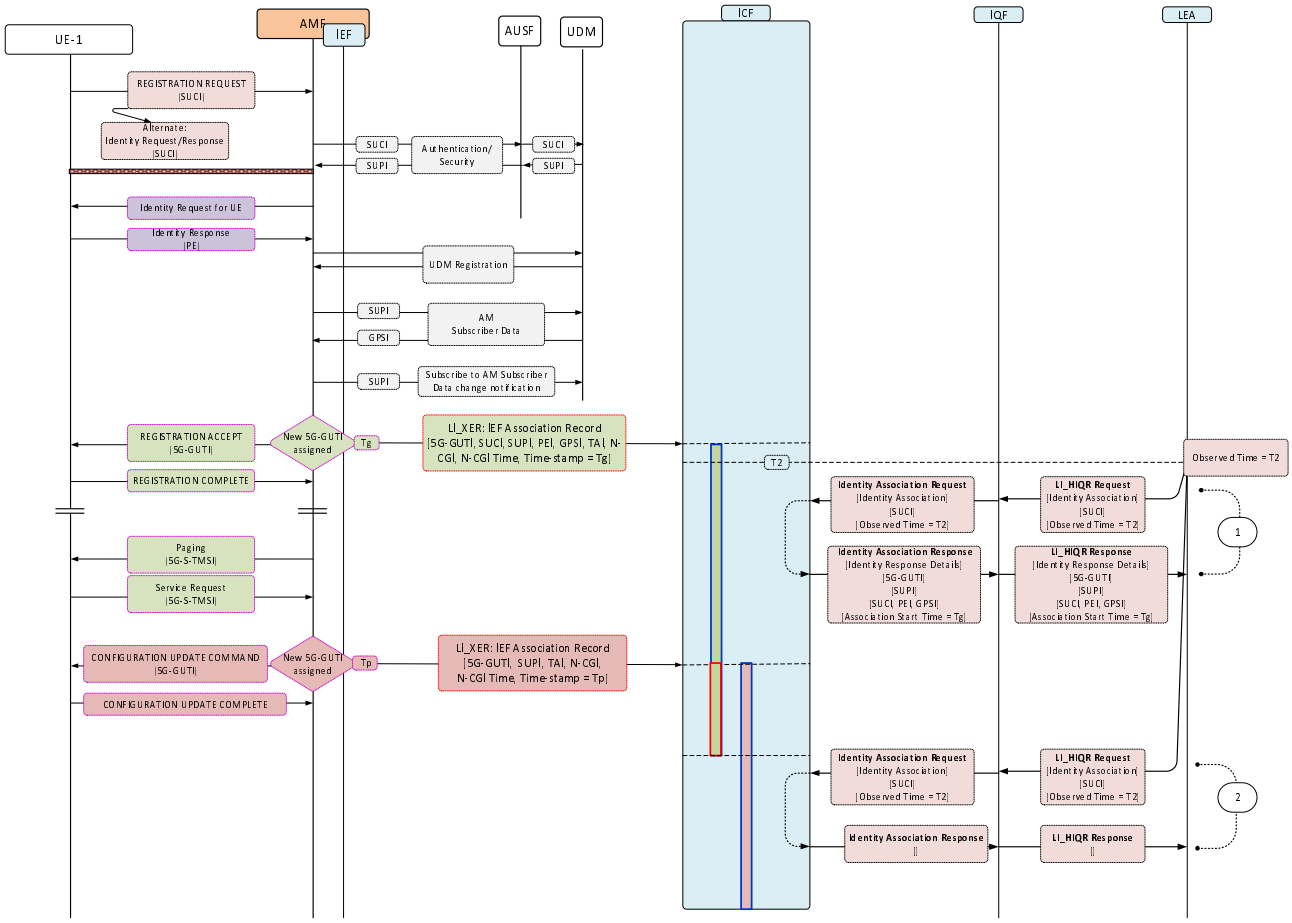


Figure 4.5.2.3-1: SUCI to SUPI retrieval – illustration 2

Figure 4.5.2.3-1 illustrates two scenarios:

Scenario 1:

- The Identity Association Request is received at the ICF while the ICF has the ID association caching for the SUCI. The Observed Time T1 in the Identity Association Request is after the Association Start Time Tg in the cached ID association. Accordingly, the Identity Association Response will have the Identity Association Record that has the Association Start Time of Tg.

Scenario 2:

- The Identity Association Request is received at the ICF after the ID association that had the SUCI was deleted from the ICF. Therefore, the Identity Association Response will have no Identity Association Records.

4.5.3 5G-S-TMSI to SUPI retrieval

4.5.3.1 General

As illustrated in the caching principles, the ID association is cached in the ICF whenever a 5G-GUTI is assigned to the SUPI.

The LEAs may observe the 5G-S-TMSI just before the next 5G-GUTI is assigned to that SUPI. To counter the time-delay in LEA submitting the request, a de-caching period is used within the ICF before deleting the previous ID association whenever a new ID association is cached.

If the LEA request with a 5G-S-TMSI is received at the ICF after the corresponding cached ID association is deleted, the response will have no Identity Association Records.

4.5.3.2 Illustration 1

Figure 4.5.3.2-1 illustrates a flow-diagram that shows the 5G-S-TMSI to SUPI retrieval handling. The flow has three scenarios based on the time when the Request is sent and the Observed Time.

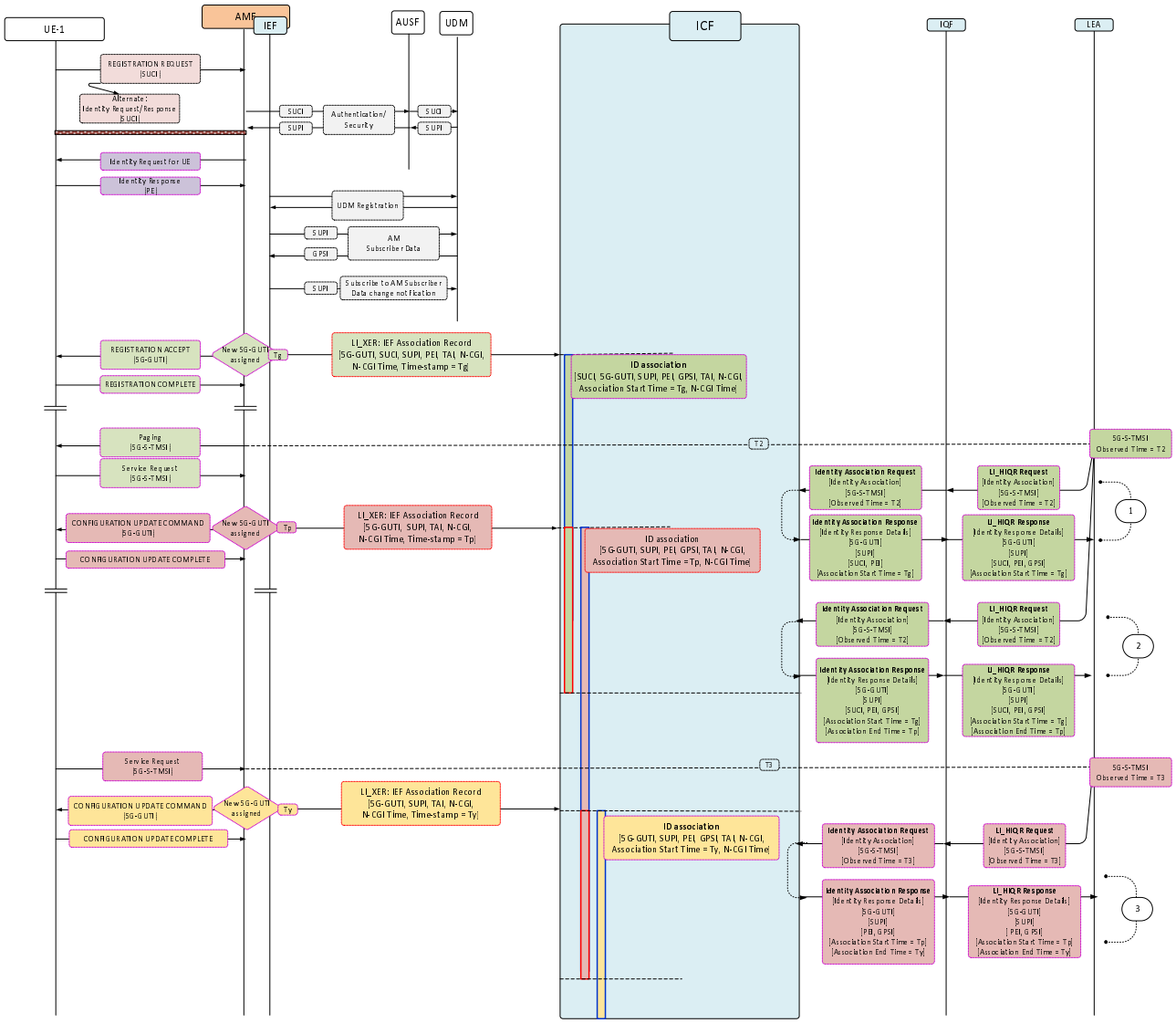


Figure 4.5.3.2-1: 5G-S-TMSI to SUPI retrieval – illustration 1

Scenario 1:

- The Identity Association Request is received at the ICF with an Observed Time of T2. When the Identity Association Request is received, the 5G-GUTI with that 5G-S-TMSI is still active as per the ID association cached in the ICF.
- The Identity Association Response will have the Identity Association Record that has the Association Start Time of Tg.

Scenario 2:

- The Identity Association Request is received at the ICF with an Observed Time of T2.
- When the Identity Association Request is received, the ID association that has the 5G-S-TMSI is in the de-caching phase and a new 5G-GUTI is assigned to the SUPI according to the cached ID association.
- The Identity Association Response will have the Identity Association Record that has the Association Start Time of Tg and Association End Time of Tp.

Scenario 3:

- The Identity Association Request is received at the ICF with an Observed Time of T3.
- When the Identity Association Request is received, the ID association that has the 5G-S-TMSI is in the de-caching phase and a new 5G-GUTI is assigned to the SUPI according to the cached ID association.
- The Identity Association Response will have the Identity Association Record that has the Association Start Time of Tp and Association End Time of Ty.

4.5.3.3 Illustration 2

Figure 4.5.3.3-1 illustrates a flow-diagram that shows the 5G-S-TMSI to SUPI retrieval handling when the two 5G-GUTIs are presumed to be associated to the SUPI (only one valid).

The flow has three scenarios based on the time the Observed Time.

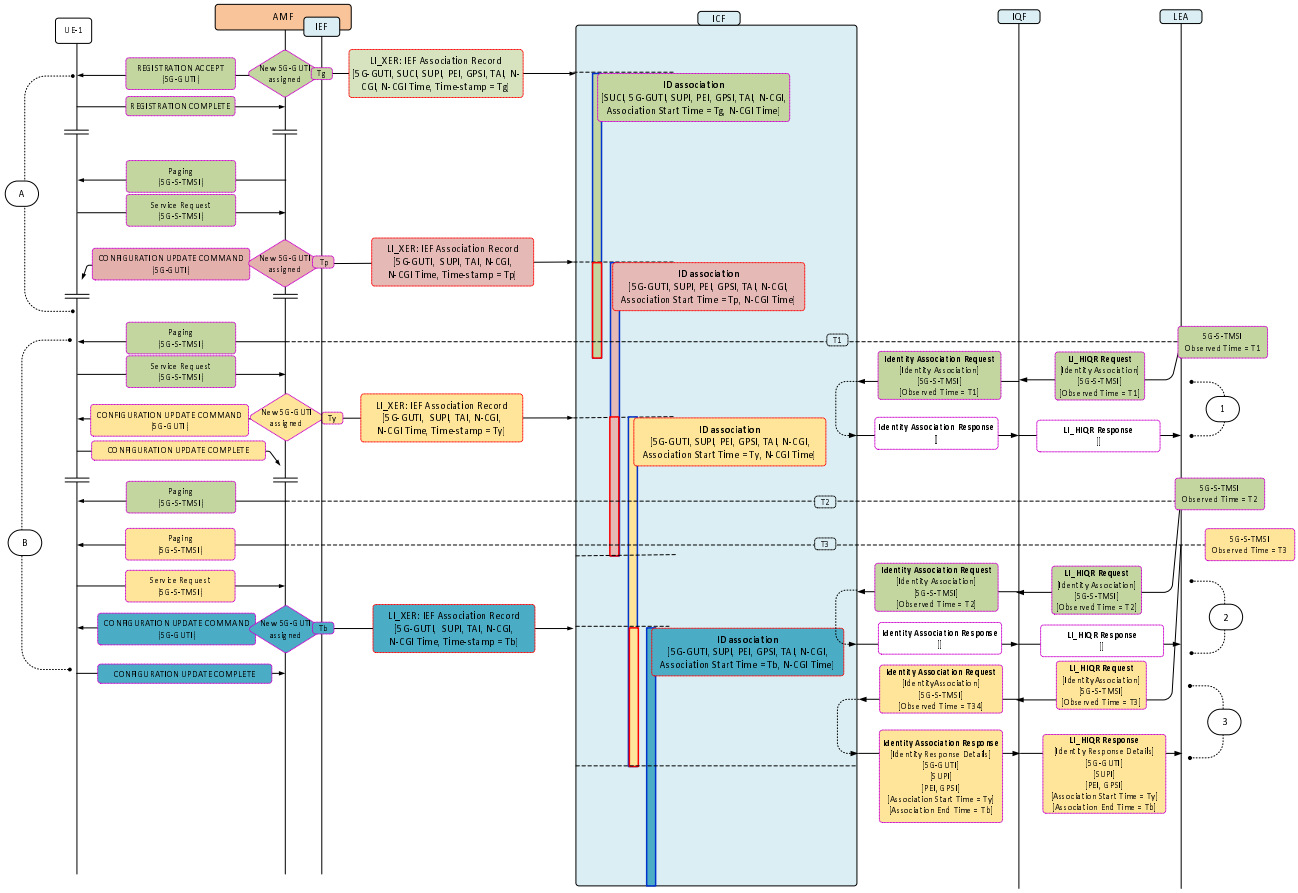


Figure 4.5.3.3-1: 5G-S-TMSI to SUPI retrieval – illustration 2

Scenario A:

- The assignment of a 5G-GUTI that has the Association Start Time of Tp (in ICF) was not successful. The CONFIGURATION UPDATE COMMAND did not reach the UE.

Scenario B:

- The assignment of a 5G-GUTI that has the Association Start Time of Ty (in ICF) was not successful. The CONFIGURATION UPDATE COMPLETE did not reach the AMF.

Scenario 1:

- The Identity Association Request is received at the ICF with an Observed Time of T1. When the Identity Association Request is received, the ID association that has the 5G-S-TMSI is already deleted from the ICF.

- Therefore, the Identity Association Response will have no Identity Association Records.

Scenario 2:

- The Identity Association Request is received at the ICF with an Observed Time of T2. When the Identity Association Request is received, the ID association that has the 5G-S-TMSI is already deleted from the ICF.
- Therefore, the Identity Association Response will have no Identity Association Records.

Scenario 3:

- The Identity Association Request is received at the ICF with an Observed Time of T3. When the Identity Association Request is received, the ID association that has the 5G-S-TMSI is in the de-caching phase and a new 5G-GUTI is assigned to the SUPI according to the cached ID association.
- The Identity Association Response will have the Identity Association Record that has the Association Start Time of Ty and Association End Time of Tb.

4.5.4 SUPI to 5G-GUTI retrieval

4.5.4.1 General

This clause illustrates the retrieval scenarios where the LI_HIQR Request has the SUPI with a Request Type of Identity Association Request.

The LEA in the LI_HIQR Request includes an Observed Time which is used within the ICF to find the active 5G-GUTI at that time.

4.5.4.2 Illustration 1

Figure 4.5.4.2-1 illustrates two scenarios based on the time at which the ICF receives the Identity Association Request. No active 5G-GUTI to SUPI association is found in the ICF at the Observed Time.

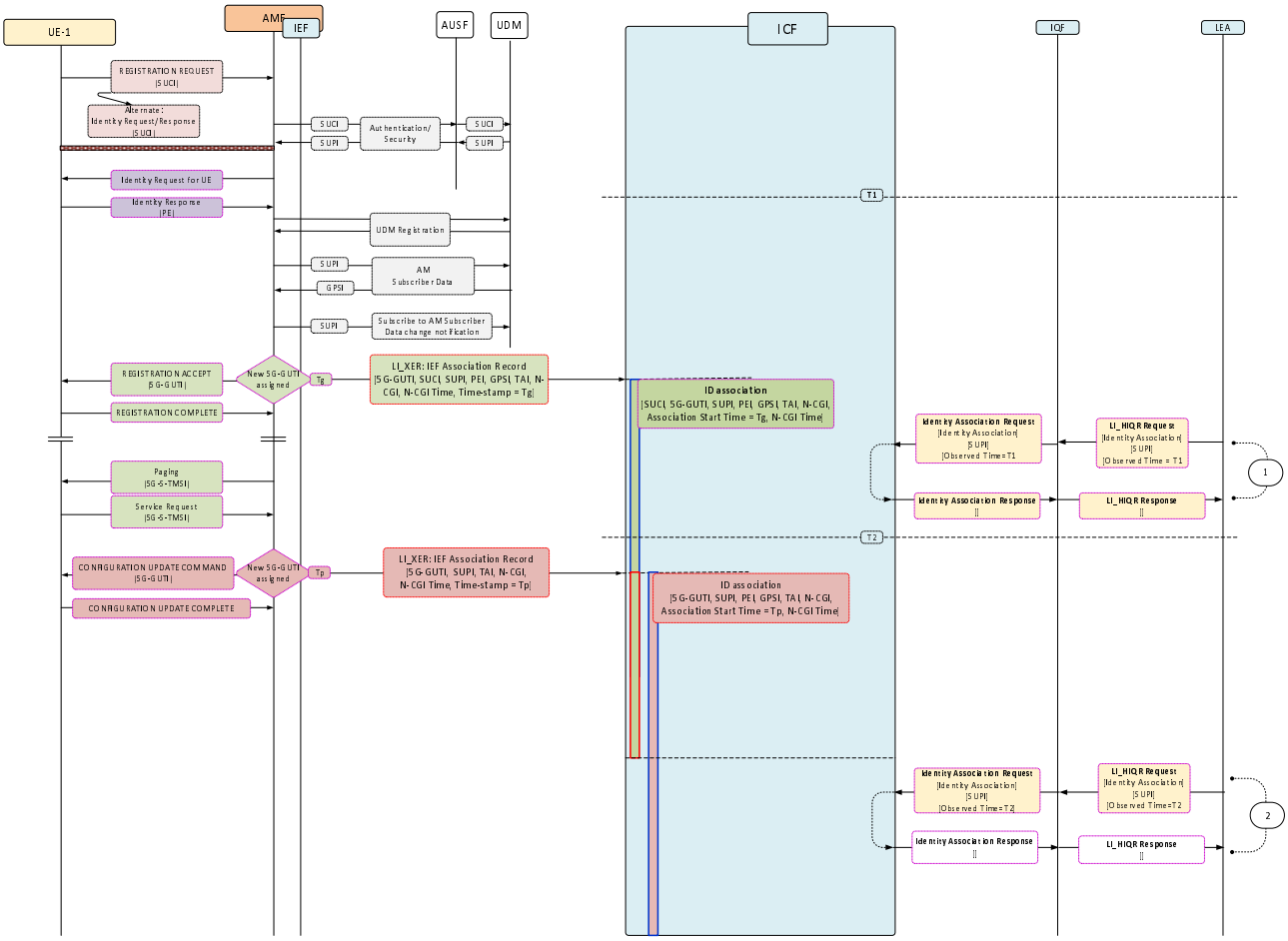


Figure 4.5.4.2-1: SUPI to 5G-GUTI retrieval – illustration 1

Scenario 1:

The Identity Association Request (with the SUPI in the Request Details) is received at the ICF with an Observed Time of T1.

The ICF has an ID association for that SUPI with an active 5G-GUTI, however, the Association Start Time Tg of that ID association is after the Observed Time of T1.

Accordingly, the Identity Association Response will have no Identity Association Records.

Scenario 2:

The Identity Association Request (with the SUPI in the Request Details) is received at the ICF with an Observed Time of T2.

The ICF has an ID association for that SUPI with an active 5G-GUTI, however, the Association Start Time Tp of that ID association is after the Observed Time of T2.

The ID association with a 5G-GUTI which was active at the Observed Time T2 was already deleted when the Identity Association Request is received at the ICF.

Accordingly, the Identity Association Response will have no Identity Association Records.

4.5.4.3 Illustration 2

Figure 4.5.4.3-1 illustrates three scenarios based on the Observed Time and the time at which the ICF receives the Identity Association Request.

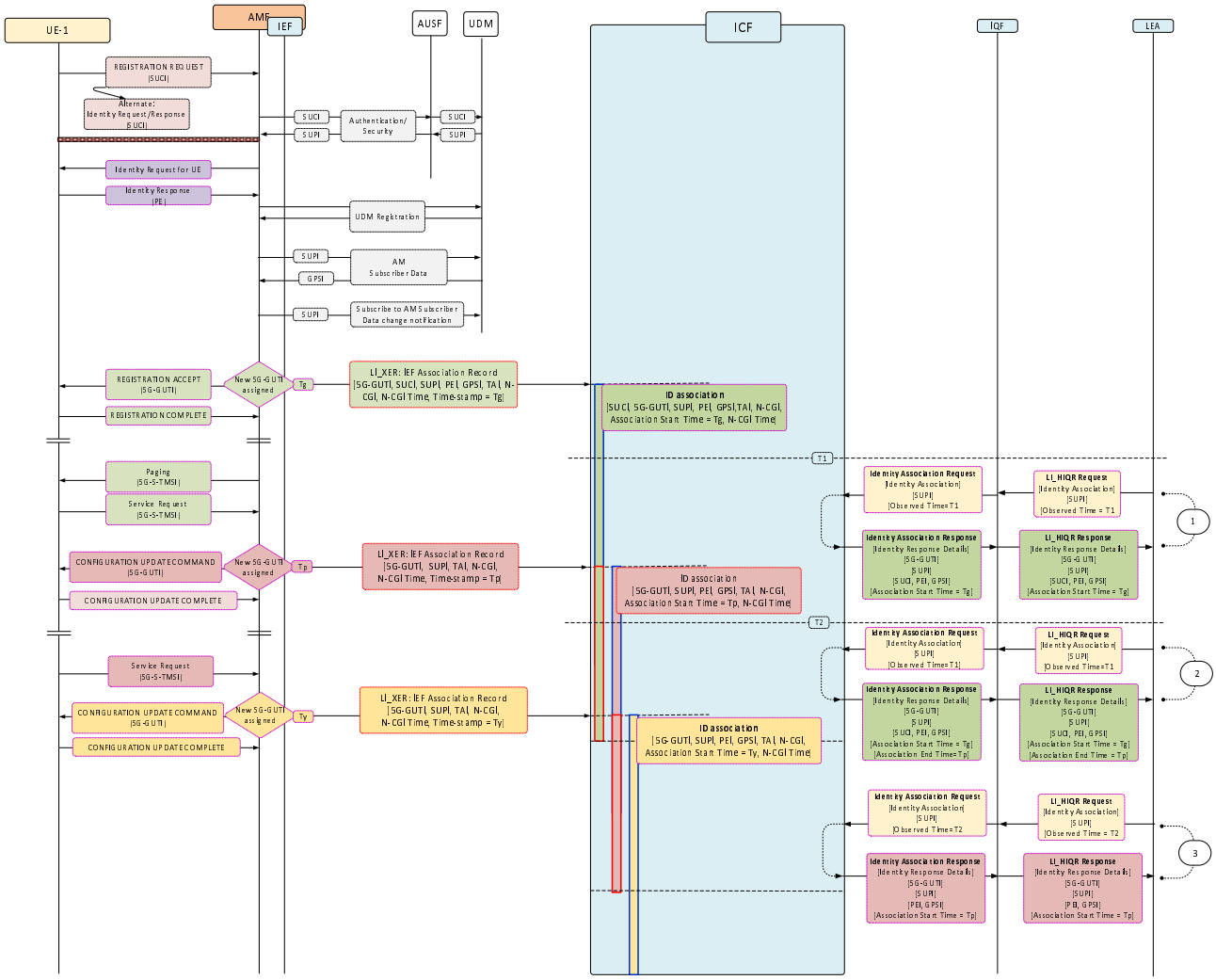


Figure 4.5.4.3-1: SUPI to 5G-GUTI retrieval – illustration 2

Scenario 1:

- The Identity Association Request (with the SUPI in the Request Details) is received at the ICF with an Observed Time of T1.
- The ICF has an ID association for that SUPI with an active 5G-GUTI, the Association Start Time Tg of that ID association is before the Observed Time of T1.
- Accordingly, the Identity Association Response will have the Identity Association Record with the 5G-GUTI (including SUCI, PEI, SUPI) with an Association Start Time of Tg.

Scenario 2:

- The Identity Association Request (with the SUPI in the Request Details) is received at the ICF with an Observed Time of T1.
- The ICF has an ID association for that SUPI with an active 5G-GUTI, however, the Association Start Time Tp of that ID association is after the Observed Time of T1.
- The ID association with a 5G-GUTI which was active at the Observed Time T1 is in de-caching phase within the ICF when the ICF receives the Identity Association Request.
- Accordingly, the Identity Association Response will have the Identity Association Record with the 5G-GUTI (including SUPI, SUCI, PEI) with an Association Start Time of Tg and Association End Time of Tp.

Scenario 3:

- The Identity Association Request (with the SUPI in the Request Details) is received at the ICF with an Observed Time of T2.
- The ICF has an ID association for that SUPI with an active 5G-GUTI, however, the Association Start Time Ty of that ID association is after the Observed Time of T2.
- The ID association with a 5G-GUTI which was active at the Observed Time T2 is in the de-caching phase within the ICF when the ICF receives the Identity Association Request.
- Accordingly, the Identity Association Response will have the Identity Association Record with the 5G-GUTI (including SUPI and PEI) with an Association Start Time of Tp and Association End Time of Ty.

4.5.4.4 Illustration 3

Figure 4.5.4.4-1 illustrates a scenario that shows the impact of a Time-Window used within the ICF for a SUPI to 5G-GUTI retrieval case.

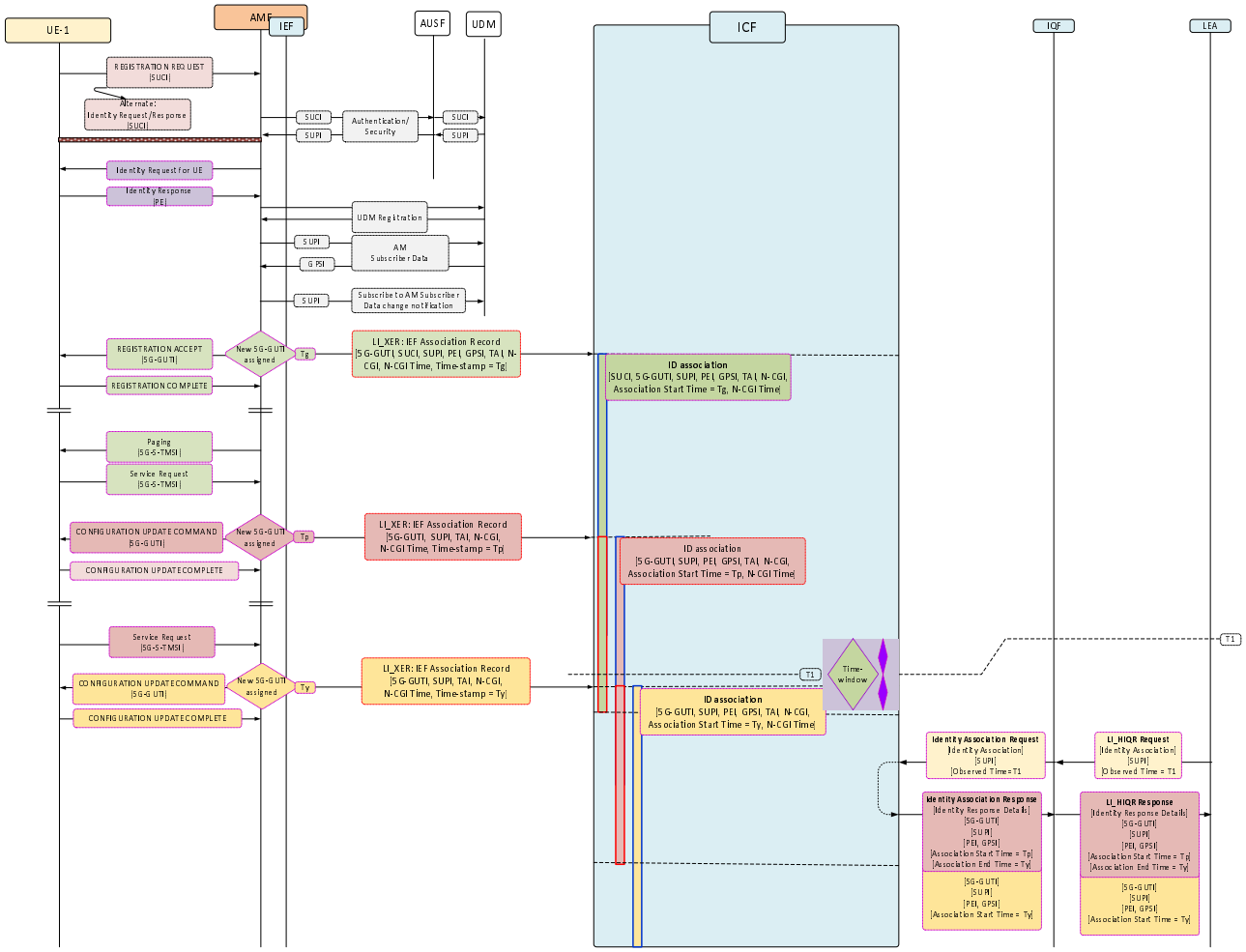


Figure 4.5.4.4-1: SUPI to 5G-GUTI retrieval – impact of Time-Window

The Identity Association Request (with SUPI in the Request Details) is received at the ICF with an Observed Time of T1.

The ICF has implemented a Time-Window concept to adjust the time-lag due to the differences between the LEA clock and the CSP clock.

The ICF has an ID association for that SUPI with an active 5G-GUTI, the Association Start Time Ty is before the Observed Time T1, if the Time-Window is applied. Without the Time-Window, the Association Start Time Ty would have been concluded to be after the Observed Time T2.

There was another ID association with a 5G-GUTI that was active at the Observed Time T2 but is in a de-caching phase within the ICF when the ICF receives the Identity Association Request.

Since two ID associations are considered active within the Time-Window of Observed Time T1, the ICF includes two Identity Association Records:

1) 5G-GUTI (also including the SUPI and the PEI) along with the Association Start Time of Tp and Association End Time of Ty.

2) 5G-GUTI (also including the SUPI and the PEI) along with the Association Start Time of Ty.

4.5.4.5 Illustration 4

Figure 4.5.4.5-1 illustrates a flow-diagram that shows the SUPI to 5G-GUTI retrieval handling when the two 5G-GUTIs are presumed to be associated to the SUPI (only one valid).

The flow has three scenarios based on the time the Observed Time.

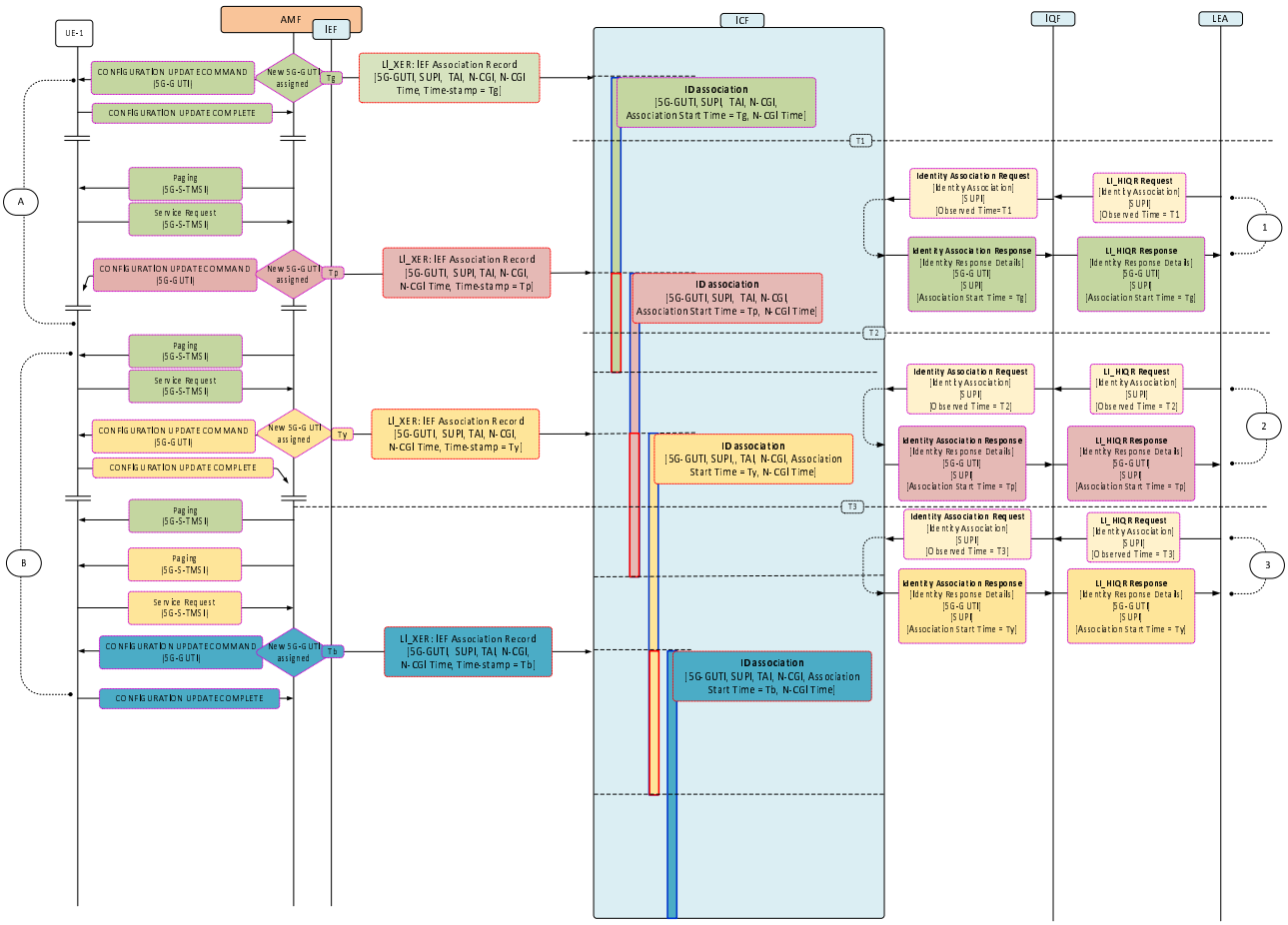


Figure 4.5.4.5-1: SUPI to 5G-GUTI retrieval - example 4

In this illustration, the record in ICF has neither PEI nor GPSI.

Scenario A:

- The assignment of a 5G-GUTI that has the Association Start Time of Tp (in ICF) was not successful. The CONFIGURATION UPDATE COMMAND did not reach the UE.

Scenario B:

- The assignment of a 5G-GUTI that has the Association Start Time of Ty (in ICF) was not successful. The CONFIGURATION UPDATE COMPLETE did not reach the AMF.

Scenario 1:

- The Identity Association Request is received at the ICF with an Observed Time of T1.
- The ICF has an ID association for that SUPI with an active 5G-GUTI and the Association Start Time Tg of that ID association is before the Observed Time of T1.
- Accordingly, the Identity Association Response will have the Identity Association Record with the 5G-GUTI (including SUPI) with an Association Start Time of Tg.

Scenario 2:

- The Identity Association Request is received at the ICF with an Observed Time of T2.
- The ICF has an ID association for that SUPI with an active 5G-GUTI and the Association Start Time Tp of that ID association is before the Observed Time of T2.
- Accordingly, the Identity Association Response will have the Identity Association Record with the 5G-GUTI (including SUPI) with an Association Start Time of Tp.

NOTE: Since the associated CONFIGURATION UPDATE COMMAND (Scenario A) never reached the UE, the 5G-GUTI included in the Identity Association Record is not the real one. But, from an ICF perspective, it is the real one.

Scenario 3

- The Identity Association Request is received at the ICF with an Observed Time of T3.
- The ICF has an ID association for that SUPI with an active 5G-GUTI and the Association Start Time Ty of that ID association is before the Observed Time of T3.
- Accordingly, the Identity Association Response will have the Identity Association Record with the 5G-GUTI (including SUPI) with an Association Start Time of Ty.

NOTE: Since the associated CONFIGURATION UPDATE COMPLETE (Scenario B) never reached the AMF, the AMF would first try to page the UE with a different 5G-GUTI. However, when that fails, the AMF would page the UE with the 5G-GUTI included in the Identity Association Record.

4.5.5 Ongoing identity association scenarios

4.5.5.1 General

This clause illustrates the end to end flow diagrams for ongoing identity association scenarios.

The LEA may send an LI_HIQR Request to activate the ongoing identity association or deactivate a previously active ongoing identity association.

4.5.5.2 Illustration 1

Figure 4.5.5.2-1 shows one example of ongoing identity association handling.

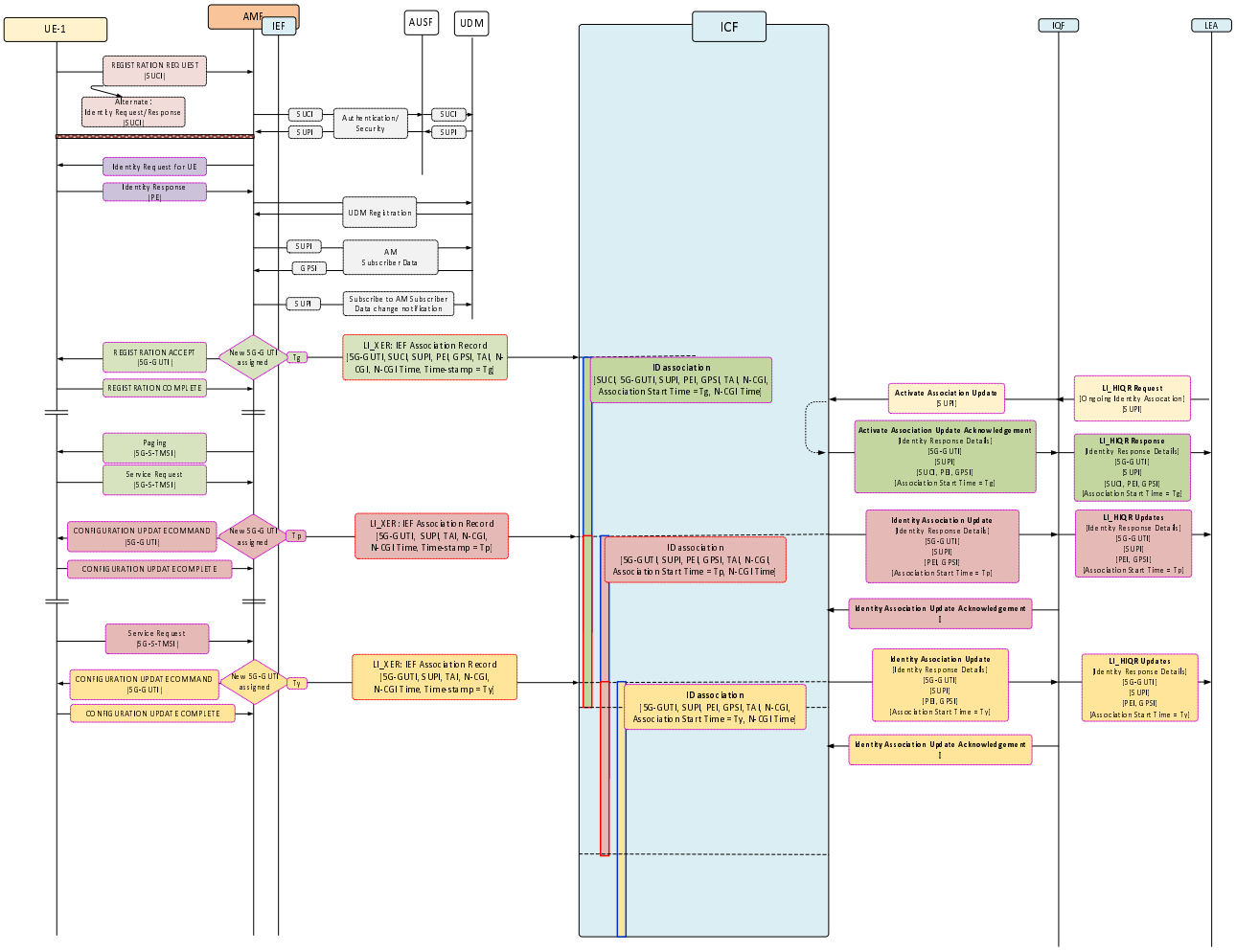


Figure 4.5.5.2-1: Ongoing identity association handling – illustration 1

In this illustration, the LEA sends an LI_HIQR Request for an ongoing identity association for a SUPI. The IQF sends an Activate Association Update message to the ICF.

The ICF finds that an active ID association is cached and includes that in the Identity Association Record of the Activate Associate Update Acknowledgement message to the IQF. The Identity Association Record will have the currently active 5G-GUTI (including the SUPI, SUCI and PEI) with an Association Start Time of Tg. The IQF forwards the received information to the LEA in the LI_HIQR Response message.

Then, a new 5G-GUTI is assigned to the UE at the AMF which is cached in the ICF with an Association Start Time of Tp. The ICF sends the Identity Association Update message to the IQF. The Identity Association Record will have the new 5G-GUTI (including the SUPI, PEI) with an Association Start Time of Tp. The IQF forwards the received information to the LEA in the LI_HIQR Updates message. The IQF would acknowledge the receipt of the Identity Association Update with an Identity Association Update Acknowledgement message.

Then, a new 5G-GUTI is assigned to the UE at the AMF which is cached in the ICF with an Association Start Time of Ty. The ICF sends the Identity Association Update message to the IQF. The Identity Association Record will have the new 5G-GUTI (including the SUPI, PEI) with an Association Start Time of Ty. The IQF forwards the received information to the LEA in the LI_HIQR Updates message. The IQF would acknowledge the receipt of Identity Association Update with an Identity Association Update Acknowledgement message.

4.5.5.3 Illustration 2

Figure 4.5.5.3-1 shows another example of ongoing identity association handling. In this example, the ICF has two ID associations, and the illustration shows that the ICF would respond with the currently active 5G-GUTI.

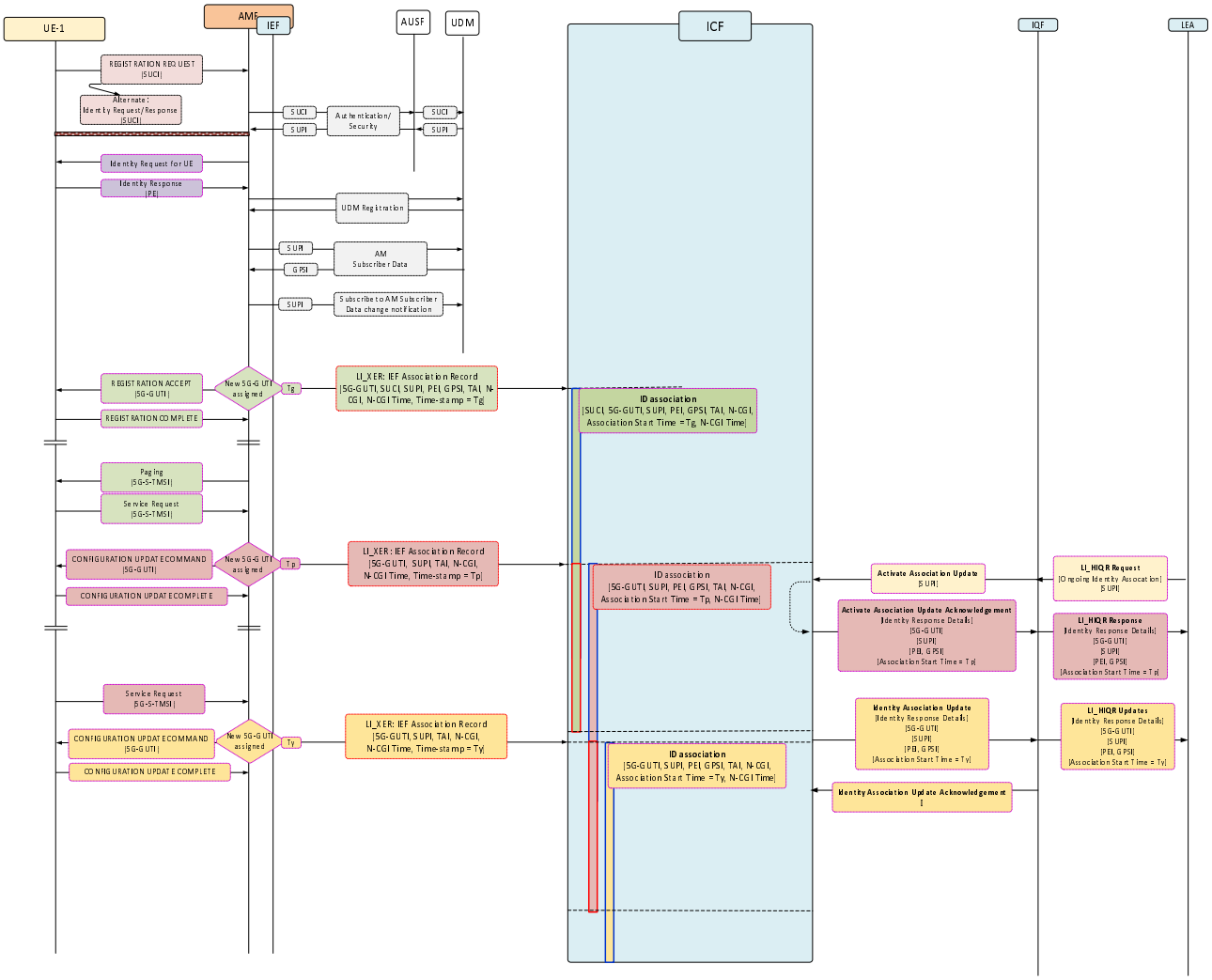


Figure 4.5.5.3-1: Ongoing identity association handling – illustration 2

In this illustration, the LEA sends an LI_HIQR Request for an ongoing identity association for a SUPI. The IQF sends an Activate Association Update message to the ICF.

The ICF finds that an ID association with the Association Start Time of Tg is marked for deletion after the de-caching period. The ICF also finds an active ID association cached and includes that in the Identity Association Record in the Activate Associate Update Acknowledgement message to the IQF. The Identity Association Record in that way will have the currently active the 5G-GUTI (also includes the SUPI and PEI) with an Association Start Time of Tp. The IQF forwards the received information to the LEA in the LI_HIQR Response message.

Then, a new 5G-GUTI is assigned to the UE at the AMF which is cached in the ICF with an Association Start Time of Ty. The ICF sends the Identity Association Update message to the IQF. The Identity Association Record will have the new 5G-GUTI (including the SUPI, PEI) with an Association Start Time of Ty. The IQF forwards the received information to the LEA in the LI_HIQR Updates message. The IQF would acknowledge the receipt of the Identity Association Update with an Identity Association Update Acknowledgement message.

4.5.5.4 Illustration 3

Figure 4.5.5.4-1 shows another example of ongoing identity association handling. In this example, the UE deregisters from the network while an ongoing identity association is active in the ICF. Subsequently, the UE registers back to the network.

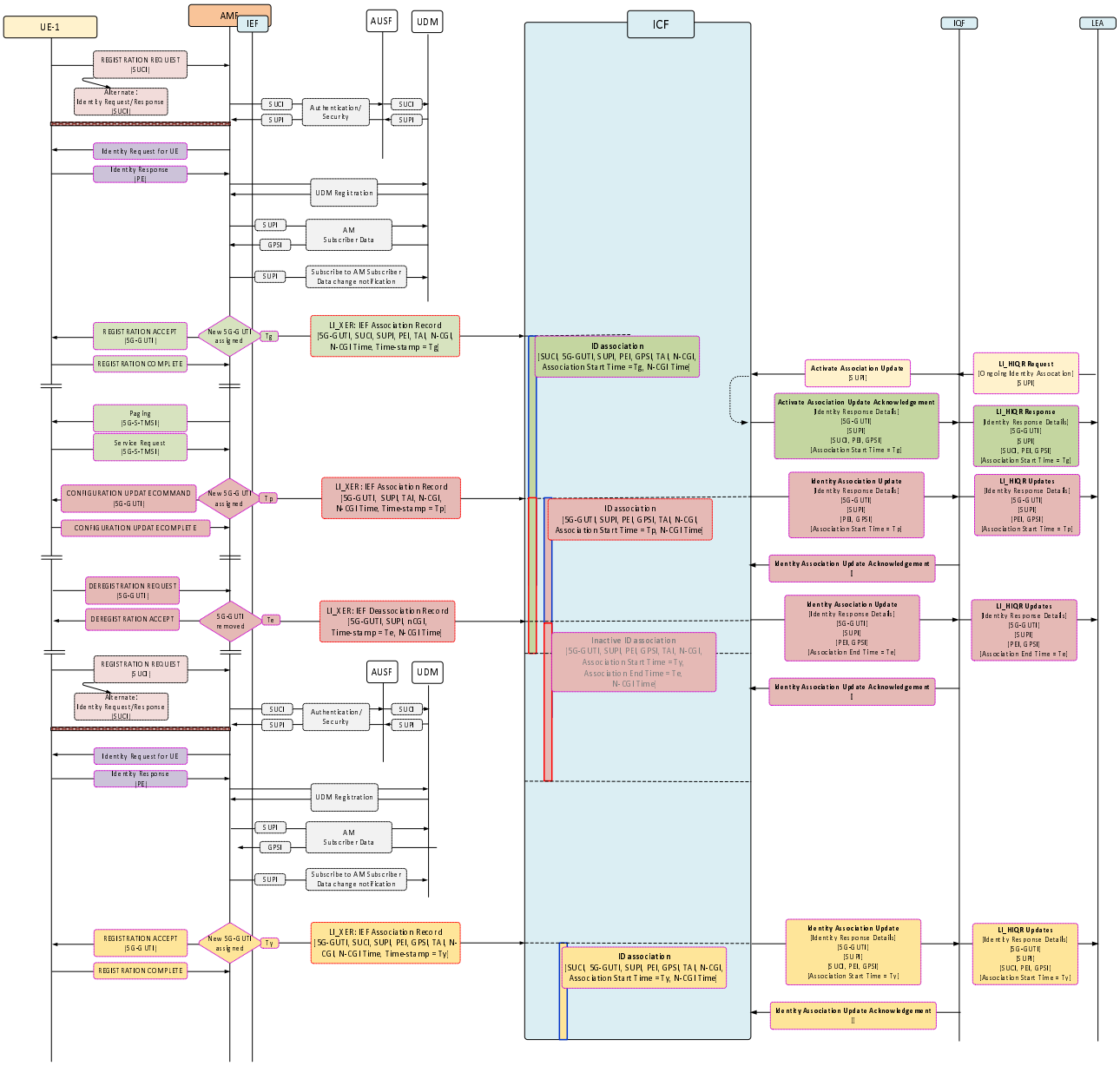


Figure 4.5.5.4-1: Ongoing identity association handling – illustration 3

In this example, the LEA sends an LI_HIQR Request for an ongoing identity association for a SUPI. The IQF sends an Activate Association Update message to the ICF.

The ICF finds that an active ID association is cached and includes that in the Identity Association Record in the Activate Associate Update Acknowledgement message to the IQF. The Identity Association Record will have the currently active 5G-GUTI (including the SUPI, SUCI and PEI) with an Association Start Time of Tg. The IQF forwards the received information to the LEA in the LI_HIQR Response message.

Then, a new 5G-GUTI is assigned to the UE at the AMF which is cached in the ICF with an Association Start Time of Tp. The ICF sends the Identity Association Update message to the IQF. The Identity Association Record will have the new 5G-GUTI (including the SUPI, PEI) with an Association Start Time of Tp. The IQF forwards the received information to the LEA in the LI_HIQR Updates message. The IQF would acknowledge the receipt of the Identity Association Update with an Identity Association Update Acknowledgement message.

Then, the UE deregisters from the network. The 5G-GUTI is removed for the SUPI in the AMF. The IEF sends an IEF Disassociation Record message to the ICF. The ICF begins the de-caching process for the corresponding ID association. The ICF also sends the Identity Association Update Message to the IQF with an Identity Association Record containing the 5G-GUTI (including the SUPI and the PEI) along with the Association End Time of Te. The IQF forwards the received information to the LEA in the LI_HIQR Updates message. The IQF would acknowledge the receipt of Identity Association Update with an Identity Association Update Acknowledgement message.

Later the UE registers to the network again. After the new 5G-GUTI is assigned, the IEF sends the IEF Association Record with the new 5G-GUTI, SUCI, SUPI, PEI to the ICF. The ICF caches the ID association with an Association Start Time of Ty. The ICF also sends the Identity Association Update Message to the IQF with an Identity Association Record containing the 5G-GUTI (including the SUPI, SUCI and the PEI) along with the Association Start Time of Ty. The IQF forwards the received information to the LEA in the LI_HIQR Updates message. The IQF would acknowledge the receipt of Identity Association Update with an Identity Association Update Acknowledgement message.

4.5.5.5 Illustration 4

Figure 4.5.5.5-1 shows another example of ongoing identity association handling. In this example, the LEA deactivates the ongoing identity association.

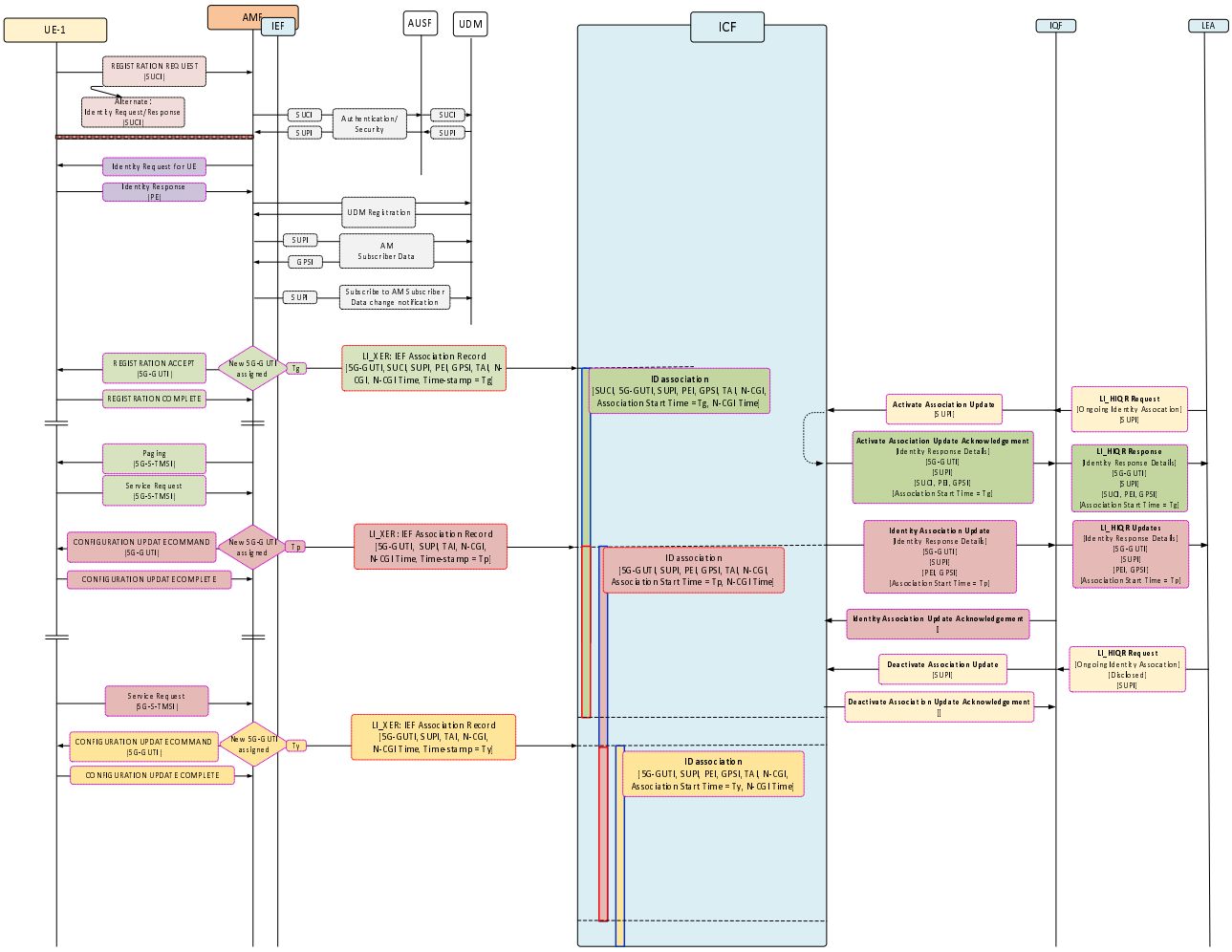


Figure 4.5.5.5-1: Ongoing identity association handling – illustration 4

In this example, the LEA sends an LI_HIQR Request for an ongoing identity association for a SUPI. The IQF sends an Activate Association Update message to the ICF.

The ICF finds that an active ID association is cached and includes that in the Identity Association Record in the Activate Associate Update Acknowledgement message to the IQF. The Identity Association Record will have the currently active 5G-GUTI (including the SUPI, SUCI and PEI) with an Association Start Time of Tg. The IQF forwards the received information to the LEA in the LI_HIQR Response message.

Then, a new 5G-GUTI is assigned to the UE at the AMF which is cached in the ICF with an Association Start Time of Tp. The ICF sends the Identity Association Update message to the IQF. The Identity Association Record will have the new 5G-GUTI (including the SUPI, PEI) with an Association Start Time of Tp. The IQF forwards the received information to the LEA in the LI_HIQR Updates message. The IQF would acknowledge the receipt of the Identity Association Update with an Identity Association Update Acknowledgement message.

Then, the LEA sends the LI_HIQR Request to deactivate the ongoing identity association for the SUPI. The IQF sends the Deactivate Association Update message to the ICF. The ICF acknowledges the receipt of Deactivate Association Update message with a Deactivate Association Update Acknowledgement message to the ICF. The ICF does not forward any message to the LEA.

Later, a new 5G-GUTI is assigned to the UE at the AMF which is cached in the ICF with an Association Start Time of Ty. The ICF does not send the Identity Association Update message to the IQF because the ongoing identity association was previously deactivated by the LEA.

4.5.5.6 Illustration 5

Figure 4.5.5.6-1 shows another example of ongoing identity association handling. In this example, the LEA deactivates the ongoing identity association while the UE was not registered to the network.

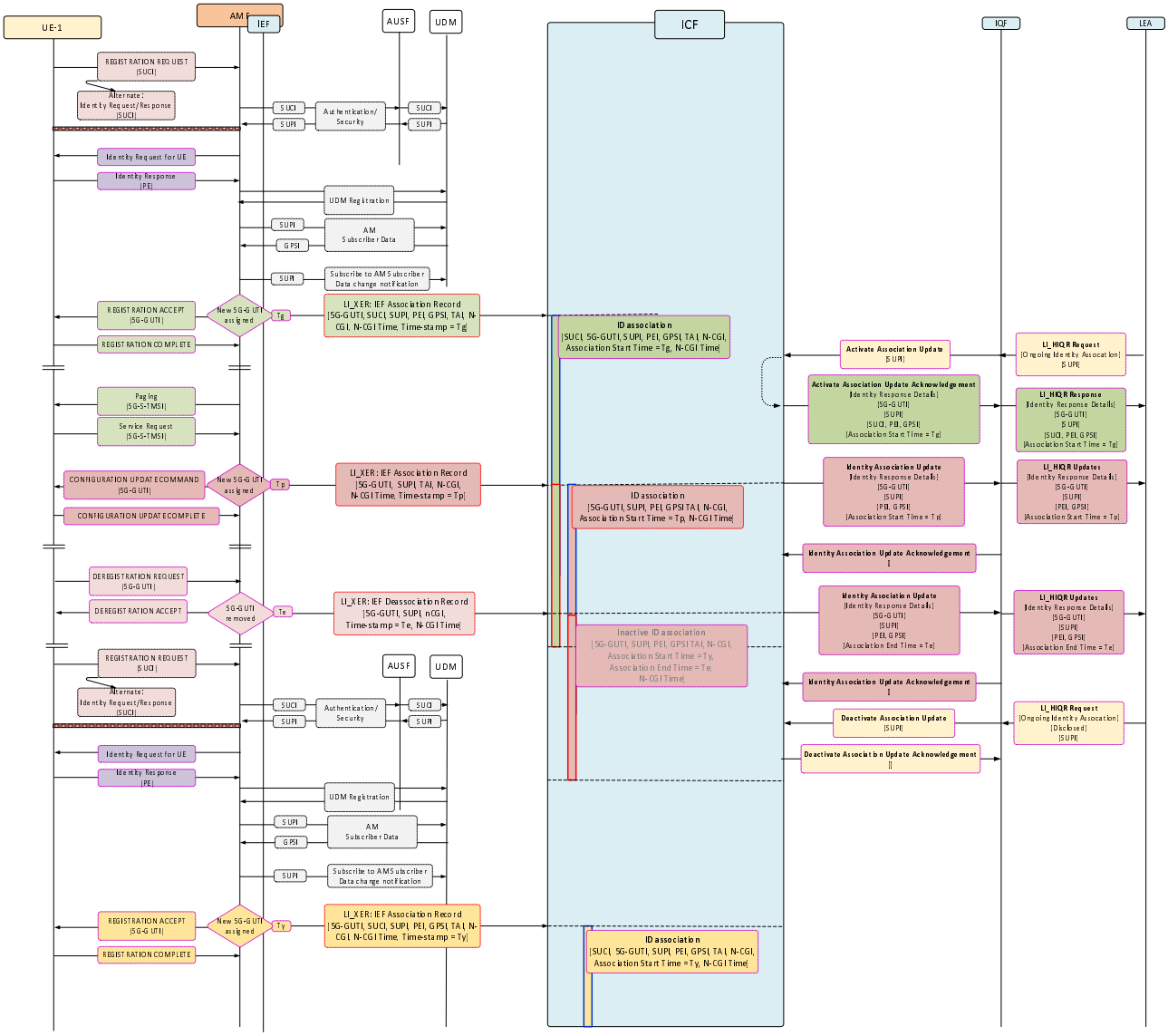


Figure 4.5.5.6-1: Ongoing identity association handling – illustration 5

In this example, the LEA sends an LI_HIQR Request for an ongoing identity association for a SUPI. The IQF sends an Activate Association Update message to the ICF.

The ICF finds that an active ID association is cached and includes that in the Identity Association Record in the Activate Associate Update Acknowledgement message to the IQF. The Identity Association Record will have the currently active 5G-GUTI (including the SUPI, SUCI and PEI) with an Association Start Time of Tg. The IQF forwards the received information to the LEA in the LI_HIQR Response message.

Then, a new 5G-GUTI is assigned to the UE at the AMF which is cached in the ICF with an Association Start Time of Tp. The ICF sends the Identity Association Update message to the IQF. The Identity Association Record will have the new 5G-GUTI (including the SUPI, PEI) with an Association Start Time of Tp. The IQF forwards the received information to the LEA in the LI_HIQR Updates message. The IQF would acknowledge the receipt of the Identity Association Update with an Identity Association Update Acknowledgement message.

Then, the UE deregisters from the network. The 5G-GUTI is removed for the SUPI in the AMF. The IEF sends an IEF Disassociation Record message to the ICF. The ICF begins the de-caching process for the corresponding ID association. The ICF also sends the Identity Association Update Message to the IQF with an Identity Association Record containing the 5G-GUTI (including the SUPI and the PEI) along with the Association End Time of Te. The IQF forwards the received information to the LEA in the LI_HIQR Updates message. The IQF would acknowledge the receipt of the Identity Association Update with an Identity Association Update Acknowledgement message.

Then, LEA sends the LI_HIQR Request to deactivate the ongoing identity association for the SUPI. The IQF sends the Deactivate Association Update message to the ICF. The ICF acknowledges the receipt of Deactivate Association Update message with a Deactivate Association Update Acknowledgement message to the ICF. The ICF does not forward any message to the LEA.

4.5.5.7 Illustration 6

Figure 4.5.5.7-1 shows another example of ongoing identity association handling. In this example, a second LEA activates the ongoing identity association and the first the LEA deactivates the ongoing identity association.

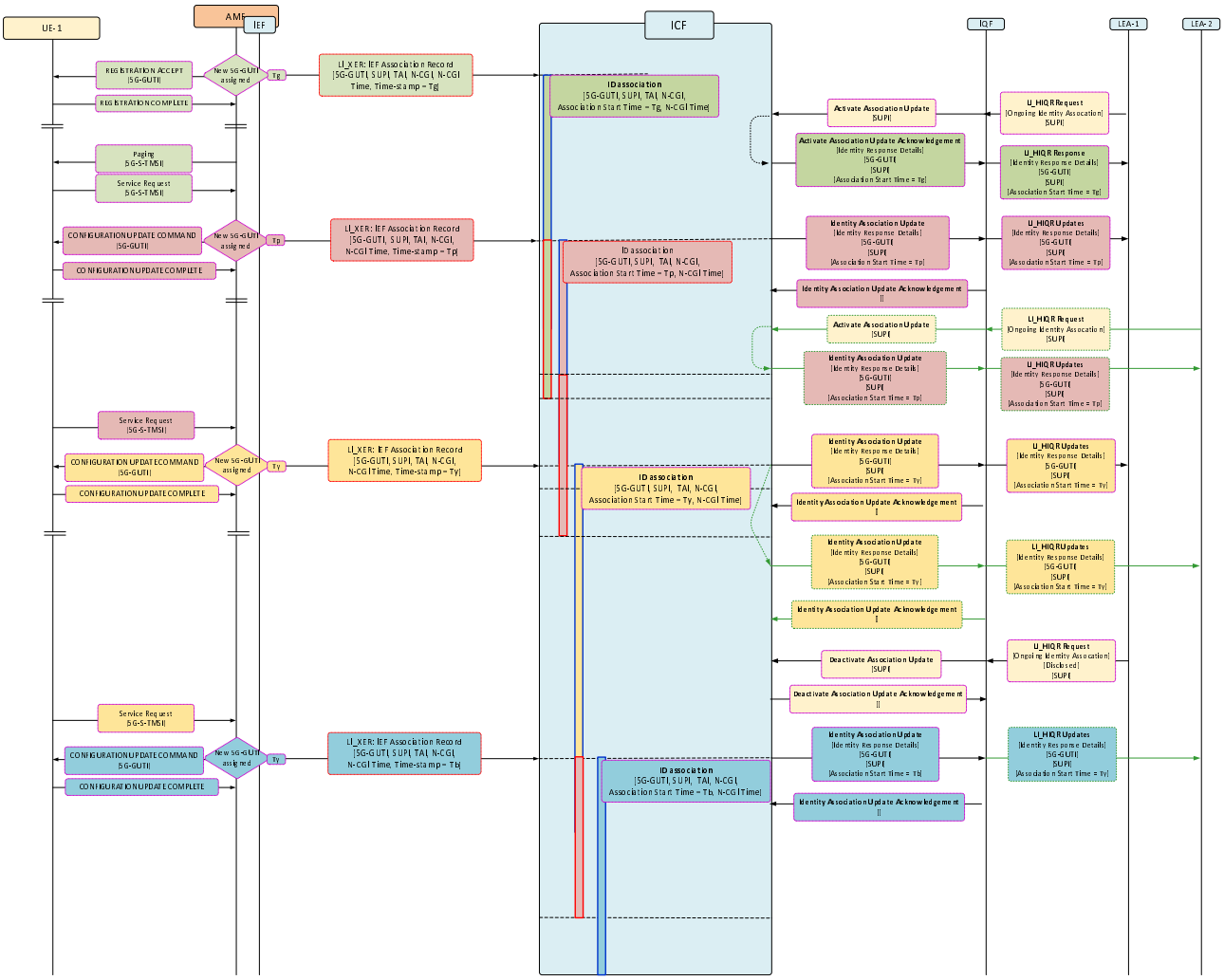


Figure 4.5.5.7-1: Ongoing identity association handling – illustration 6

In this example, the LEA-1 sends an LI_HIQR Request for an ongoing identity association for a SUPI. The IQF sends an Activate Association Update message to the ICF. In the illustration, the ICF records have neither the PEI nor the GPSI.

The ICF finds that an active ID association is cached and includes that in the Identity Association Record in the Activate Associate Update Acknowledgement message to the IQF. The Identity Association Record will have the currently active 5G-GUTI (including the SUPI, SUCI and PEI) with an Association Start Time of Tg. The IQF forwards the received information to the LEA in the LI_HIQR Response message.

Then, a new 5G-GUTI is assigned to the UE at the AMF which is cached in the ICF with an Association Start Time of Tp. The ICF sends the Identity Association Update message to the IQF. The Identity Association Record will have the new the 5G-GUTI (including the SUPI, PEI) with an Association Start Time of Tp. The IQF forwards the received information to the LEA in the LI_HIQR Updates message. The IQF would acknowledge the receipt of the Identity Association Update with an Identity Association Update Acknowledgement message.

The LEA-2 sends an LI_HIQR Request for an ongoing identity association for a SUPI. The IQF sends an Activate Association Update message to the ICF.

The ICF finds that an active ID association is cached and includes that in the Identity Association Record in the Activate Associate Update Acknowledgement message to the IQF. The Identity Association Record will have the currently active the 5G-GUTI (including the SUPI, SUCI and PEI) with an Association Start Time of Tp. The IQF forwards the received information to the LEA in the LI_HIQR Response message.

Then, a new 5G-GUTI is assigned to the UE at the AMF which is cached in the ICF with an Association Start Time of Ty. The ICF sends the Identity Association Update message to the IQF. The Identity Association Record will have the new 5G-GUTI (including the SUPI, PEI) with an Association Start Time of Ty. The IQF forwards the received information to the LEA-1 in the LI_HIQR Updates message. The IQF would acknowledge the receipt of the Identity Association Update with an Identity Association Update Acknowledgement message.

The ICF sends another Identity Association Update message to the IQF. The Identity Association Record will have the new 5G-GUTI (including the SUPI, PEI) with an Association Start Time of Ty. The IQF forwards the received information to the LEA-2 in the LI_HIQR Updates message. The IQF would acknowledge the receipt of the Identity Association Update with an Identity Association Update Acknowledgement message.

Then, LEA-1 sends an LI_HIQR Request to deactivate the ongoing identity association for the SUPI. The IQF sends the Deactivate Association Update message to the ICF. The ICF acknowledges the receipt of Deactivate Association Update message with a Deactivate Association Update Acknowledgement message to the ICF. The ICF does not forward any messages to the LEA.

Later, a new 5G-GUTI is assigned to the UE at the AMF which is cached in the ICF with an Association Start Time of Tb. The ICF sends another Identity Association Update message to the IQF. The Identity Association Record will have the new 5G-GUTI (including the SUPI, PEI) with an Association Start Time of Tb. The IQF forwards the received information to the LEA-2 in the LI_HIQR Updates message. The IQF would acknowledge the receipt of the Identity Association Update with an Identity Association Update Acknowledgement message.

Annex A:
Change history

Change history							
Date	**Meeting**	**TDoc**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New version**
2024-07	SA3LI#94	S3i240508				Initial version	0.0.1
2025-01	SA3LI#96	S3i250013				Inclusion of pCR in S3i250010	0.0.2
2026-07	SA3LI#102	S3i260360				Agreed for Change Control	0.0.3
2026-08	SA#113	SP-260731				Submitted to SA#113 Plenary for Approval	1.0.0
2026-09	SA#113	-				Approved by SA#113 Plenary for Change Control	19.0.0

History

Version	Date	Status
V19.0.0	October 2026	Publication